Financial Conditions, r^*, and the Transmission of Monetary Policy Surprises

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Abstract

Interest-rate surprises around Federal Reserve announcements are a clean measure of monetary policy shocks. But their estimated effects on unemployment and inflation since 1994 are often weak or have the wrong sign. We keep the surprise as it is and change only how its effects are traced through the economy. If the surprise is traced through the two-year yield alone, the wrong signs appear. If it is traced through financial conditions as a whole, measured relative to the neutral interest rate r^*, the effects have the expected signs and are precisely estimated. A policy move that raises the two-year yield 1 percentage point raises the unemployment rate about 0.4 percentage point and lowers industrial production about 1 percent within a year. It lowers core prices about 0.6 percent after two years. Including financial conditions matters most for the unemployment result. Measuring them relative to r^* matters most for the price result, which relies on data since 2020.

1 Introduction

How does monetary policy affect the economy? Standard theory says that a surprise tightening should raise unemployment and lower inflation. Yet empirical work often finds the opposite signs, or effects too imprecise to tell. The problem is worst in samples that start after the mid-1980s, when policy followed the economy more closely and left fewer clear shocks to learn from (Ramey, 2016).

Interest-rate surprises around Federal Reserve announcements were meant to solve this. The FOMC makes its decision before the announcement, so market moves in a narrow window around the announcement cannot have caused the decision. The change in interest-rate futures and Treasury yields in that window is then news about policy (Kuttner, 2001; Gürkaynak et al., 2005). These surprises measure the effects of policy on asset prices precisely (Bernanke and Kuttner, 2005; Hanson and Stein, 2015; Nakamura and Steinsson, 2018; Swanson, 2021). But when they are used to measure effects on unemployment and inflation, the old puzzles often return.

Recent work offers three explanations, each with a fix.

- *Information.* An announcement reveals the Fed's view of the economy as well as its policy. A surprise tightening can then signal good news about the economy (Romer and Romer, 2000; Nakamura and Steinsson, 2018). Jarociński and Karadi (2020) separate the two using a simple idea: a pure policy tightening should lower stock prices, so when rates and stocks rise together, the surprise mostly carries information. The pure policy part gives a larger fall in prices (see also Miranda-Agrippino and Ricco, 2021).
- *Predictable surprises.* Part of each surprise can be predicted from public data released before the announcement, so it is not a clean shock. And after 1984 there are few large surprises to

learn from. Bauer and Swanson (2023a) add surprises from speeches by the Fed Chair and remove the predictable part. Their effects on the economy are larger and more precise (see also Bauer and Swanson, 2023b).

- *Timing.* Markets react within minutes, but the economist sees unemployment and inflation only as monthly averages. Jacobson et al. (2026) show that this mismatch alone can produce wrong signs. When they measure inflation daily, using online prices from 2008 to 2015 and keeping the same surprises and the same method, most of the wrong-signed response disappears.

The first two fixes change the surprise. The third changes how the outcome is measured. Gertler and Karadi (2015) and Caldara and Herbst (2019) add a fourth point: credit costs carry part of the effect, and leaving out credit spreads weakens it.

This note offers a new take. We keep the surprise as it is, and we keep the outcomes and the method fixed. We change only how we measure the variables that carry policy to the economy. Standard models carry the surprise forward through one interest rate, measured in nominal terms. That leaves out two basic parts of how monetary policy works.

First, policy affects spending through financial conditions as a whole: the cost of long-term credit, the value of wealth and the exchange rate (Boivin et al., 2010). A surprise tightening moves long-term yields, stock prices and the dollar, not only the two-year yield. Policy analysts often summarize these conditions in a financial conditions index (Hatzius et al., 2010; Ajello et al., 2023). Hatzius et al. (2010) make the same point: financial conditions respond to policy but also to other forces, so they can diverge from the policy rate, as in 2004–06, when conditions stayed easy while the funds rate rose.

Second, what matters for spending and inflation is the level of real interest rates compared with the neutral rate r^*. The neutral rate is the rate that neither speeds up nor slows down the economy. It fell a lot from the 1990s to the 2010s (Laubach and Williams, 2003; Holston et al., 2017, 2023; Hamilton et al., 2016; Kiley, 2020b), and it shaped the long-run trend in yields (Bauer and Rudebusch, 2020). Since 1994, interest rates have trended down along with r^*, while activity has trended up. A model that links the level of rates to activity then partly picks up these opposite trends. It finds that higher rates go with stronger activity, which looks like a wrong-signed effect of policy.

The two parts need not matter equally for every outcome. Activity responds to changes in financial conditions. Inflation depends on how long real rates stay above or below r^*, through the slack this creates. So measuring rates relative to r^* should matter most for prices.

Our design is simple. The shock is the policy-news surprise of Nakamura and Steinsson (2018), measured in narrow windows around 252 FOMC announcements from 1996 to 2026. We read its effect on the two-year yield, and on a market-based financial conditions index (FCI), from these windows. We assume that unemployment and inflation do not move within the windows. A daily model carries the two-year yield and the FCI forward from the announcement day. Monthly unemployment and core inflation then respond to the monthly averages of these daily series, as in a standard monthly model (a vector autoregression, or VAR). Like Jacobson et al. (2026), we take the timing of each announcement within its month into account. The FCI and the order of steps—nominal, real, then relative to r^*—come from a companion note on financial conditions (FCI note, 2026). We fixed them before this analysis.

The evidence builds in four steps. Every step includes a smooth time trend (a quadratic) in the monthly equations.

- With the nominal two-year yield as the only channel, the post-1994 puzzles appear. The unemployment response changes with the number of past months in the model (lags). With

12 lags it is negative, and its 90 percent interval excludes zero. Below, "significant" means that the 90 percent interval excludes zero. Prices barely move, and industrial production rises.

- Adding the FCI makes the unemployment response standard, large and significant. But prices fall only a little, and industrial production does not fall.
- Measuring rates in real terms changes little.
- Measuring the two-year yield and the FCI relative to r^* completes the picture. Unemployment rises 0.36 to 0.40 percentage point after 12 months. The core price level falls 0.53 to 0.72 percent after 24 months. Industrial production falls about 1 percent after 12 months. All are significant for every number of lags.

Compared with the benchmark of Bauer and Swanson (2023a), our price effect is similar for two years and larger after that. Our effect on industrial production is somewhat smaller. Our effect on unemployment is about twice as large after one year and lasts longer.

The robustness checks show what drives these results. Financial conditions are essential for unemployment. Without the FCI, no adjustment of the two-year yield gives a reliable unemployment response. With the FCI, every adjustment does. Measuring conditions relative to r^* is essential for prices. The price effect is strong whenever the FCI is measured relative to the Laubach–Williams r^*, however the two-year yield is adjusted. The unemployment response holds up however we treat the pandemic, and it is larger in a sample that ends in 2019. The significant price effect, by contrast, depends mainly on data since 2020. The results are the same or slightly stronger with the cleaned surprises of Bauer and Swanson (2023a) and the pure-policy surprises of Jarociński and Karadi (2020). They also hold with a fuller method for computing uncertainty.

1.1 Connection to the literature

Our paper is closest in spirit to Jacobson et al. (2026). Like them, we hold the surprise and the method fixed and change one measurement. They change how often the outcome is measured. We change how the channel from policy to the economy is measured. They study inflation only; they do not estimate effects on activity, which is where our main result lies. Both changes remove wrong-signed responses, and both point to a mismatch between what the economist measures and what drives decisions: daily rather than monthly information in their case, financial conditions relative to r^* rather than a single nominal rate in ours. Our model also uses the timing of announcements within the month, which is central to their argument. Related work on timing includes Kilian (2024), Chudik and Georgiadis (2022), Francis et al. (2011), Ghysels (2016) and Gazzani and Vicondoa (2020).

Bauer and Swanson (2023a) and Jarociński and Karadi (2020) improve the surprise itself. In our design, a surprise enters only through its immediate effect on the two-year yield and the FCI, and through its timing. Using their cleaned surprises in our model changes the results little and makes them slightly stronger. So their fixes and ours work in the same direction and can be combined. Our approach also changes which assumptions the estimates rest on. It depends less on how strongly the surprise moves rates over a month, and less on its link to earlier news. But it relies on a different assumption: the economy responds to a given change in financial conditions in the same way, whatever caused the change. Gertler and Karadi (2015) and Caldara and Herbst (2019) are closest in emphasizing financial channels; we use a broad index and measure it relative to r^*, but we leave out credit spreads. The use of futures-based surprises in monthly models goes back to Faust et al. (2004), and the choice of surprise series can matter for the results (Bu et al.,

2021; Brennan et al., 2025). Caballero et al. (2026) make a related point about the neutral level: because r^* must absorb asset-price movements that a financial conditions index already captures, they propose a neutral level of the index itself, FCI*, estimated in a Laubach–Williams-type model for 1990–2024 with FCI-G. We keep r^* for the rates inside the two-year yield and the index and do not estimate FCI*; using gaps to FCI* instead is a natural extension. To our knowledge, no earlier study traces the effects of these surprises through a broad financial conditions index measured relative to r^*.

2 Background: how policy reaches the economy

Before turning to the data, we set out the reasoning behind this design. We first describe how monetary policy is thought to reach spending. We then show why a standard VAR can miss this route, and why that could produce the puzzles. Finally, we connect the argument to work on financial conditions and economic activity.

2.1 The standard transmission mechanism

Monetary policy sets the short-term interest rate today and, through its communications, shapes what people expect that rate to be in the future. But few households and businesses borrow at the policy rate. What they face are the rates on consumer credit, mortgages and business debt. Arbitrage links these rates to current and expected short-term rates. So when the expected path of policy rises, the rates that households and businesses pay rise too, and spending falls.

The same rise in interest rates also raises the return that investors require on stocks. For businesses, the costs of debt and equity together make up the cost of capital. A higher cost of capital raises the user cost of new equipment and buildings, and investment falls (Jorgenson, 1963; Hall and Jorgenson, 1967). The value of existing capital relative to its replacement cost falls as well (Tobin, 1969).

For households, a higher required return lowers stock prices and so lowers wealth. Consumption depends on wealth as well as on current income (Ando and Modigliani, 1963; Friedman, 1957), so spending falls. Higher interest rates pass through to the value of other assets, such as houses, in a similar way.

Higher interest rates also strengthen the dollar. A stronger dollar shifts spending away from goods produced at home and toward imports, and it reduces exports (Fleming, 1962; Mundell, 1963).

Interest rates of many maturities, equity prices and the dollar therefore form the core of the transmission mechanism (Boivin et al., 2010). These links run from the early work cited above to modern models of the business cycle, in closed and open economies (Christiano et al., 2005; Smets and Wouters, 2007; Galí and Monacelli, 2005). Models with heterogeneous households change the relative importance of the links, giving more weight to the indirect effects that work through income (Kaplan et al., 2018; Auclert, 2019). But in all of these models the short-term rate matters mainly because of what it does to the prices and rates that people actually face.

What matters for spending in these models is not the nominal rate, but the real rate relative to the neutral rate r^*. The same holds for the other parts of financial conditions. A given yield or stock valuation is tighter when r^* is low than when it is high.

2.2 What standard VARs leave out

The workhorse evidence on monetary policy comes from VARs with a few macroeconomic variables and one interest rate (Bernanke and Blinder, 1992; Christiano et al., 1999). Some studies add a credit spread (Gertler and Karadi, 2015; Caldara and Herbst, 2019), and factor models bring in many more series (Bernanke et al., 2005). But long-term yields, equity valuations and the dollar are usually absent from the small models used to measure the effects of policy. These are the variables that the mechanism above says carry those effects. Concern about missing variables is not new: Sims (1992) argued that the price puzzle arises because the Federal Reserve responds to information that small VARs do not contain. Our concern is different. Here the missing variables are the channel itself.

A simple model shows what this omission can do. Let U_t be the unemployment gap, i_t the policy rate and f_t financial conditions, with a higher f_t meaning tighter conditions:

$$U_t = \rho_U \, U_{t-1} + \kappa \, f_{t-1} + \varepsilon_t^U , \tag{1}$$

$$f_t = \rho_f \, f_{t-1} + \lambda \, i_t + \varepsilon_t^f . \tag{2}$$

Unemployment responds to financial conditions and not directly to the policy rate. Financial conditions respond to the policy rate and to other forces, ε_t^f, such as changes in risk appetite. Suppose the economist observes only U_t and i_t. To remove f_t, lag equation (1) one month, multiply it by ρ_f, subtract it from equation (1), and use equation (2). This gives

$$U_t = (\rho_U + \rho_f) \, U_{t-1} - \rho_U \rho_f \, U_{t-2} + \kappa\lambda \, i_{t-1} + \underbrace{\varepsilon_t^U - \rho_f \, \varepsilon_{t-1}^U + \kappa \, \varepsilon_{t-1}^f}_{\text{error}} . \tag{3}$$

The policy rate has no direct effect in this model. It appears in equation (3) because it stands in for financial conditions. The error is a moving average that includes past financial shocks, so a VAR in U_t and i_t with few lags cannot capture it. The VAR is misspecified. With the lags and samples available in practice, this can make the effects of monetary policy harder to measure. In some settings a policy shock may not be recoverable at all from a small set of variables (Fernández-Villaverde et al., 2007), but our argument does not depend on that.

There is a second problem, and it matters more for the size and sign of the response: the omission changes the experiment being traced. In the announcement windows, the financial conditions index used below rises about 1 unit per percentage point rise in the two-year yield. In ordinary months, its unexpected movements are only about 0.35 to 0.4 unit per percentage point. A VAR without financial conditions traces the typical move in the two-year yield, which comes with a much smaller change in financial conditions than a policy surprise does.

A small simulation shows the size of the problem. The true model is a monthly VAR in the unemployment gap, core PCE inflation, the two-year yield and the financial conditions index, with six lags and a quadratic trend. Macroeconomic variables are ordered first. We estimate it on the sample described in Section 3, which also defines the index and its nominal and r^*-gap versions. We generate 2,000 artificial samples of the same length (382 months) from this model. On each sample, we estimate the VAR with all four variables and again without the index. We then trace a policy move that raises the two-year yield 1 percentage point, with no immediate effect on unemployment or inflation. With the index in the VAR, the move also raises the index by 0.99 unit, the amount measured in the announcement windows. Without it, the same move raises only the yield. Table 1 shows the results. These VARs are monthly and simpler than the model of Section 3, so their true responses are not the estimates reported in Section 4.

Table 1: Simulation: estimating the effects of policy with and without financial conditions

		True	With the FCI		Without the FCI	
		response	Mean (s.d.)	Wrong sign	Mean (s.d.)	Wrong sign
Nominal	Unemployment, 12 months	+0.36	+0.32 (0.18)	4%	-0.08 (0.18)	66%
	Price level, 24 months	-0.32	-0.34 (0.25)	8%	-0.21 (0.23)	18%
r^*-gap	Unemployment, 12 months	+0.51	+0.43 (0.19)	1%	+0.11 (0.17)	26%
	Price level, 24 months	-0.84	-0.75 (0.25)	0%	-0.57 (0.22)	1%

Responses to a policy move that raises the two-year yield 1 percentage point; unemployment gap in percentage points, core PCE price level in percent. True model: monthly VAR in the unemployment gap, core PCE inflation, the two-year yield and the FCI (six lags, quadratic trend), estimated on 1994–2026, with the two-year yield and the FCI in the form shown. 2,000 samples of 382 months; estimated VARs have six lags. Mean and s.d.: across samples. "Wrong sign": share of samples in which the estimate has the opposite sign to the true response.

The VAR with the index comes close to the true responses. The VAR without it does not. In the nominal case, the true unemployment response after 12 months is +0.36. The VAR without the index gives −0.08 on average, and the estimate has the wrong sign in two-thirds of the samples. In the r^*-gap case, it gives +0.11 against a true response of +0.51. The price effects are also understated, though less severely. More data would reduce the noise but not this error: with ten times as many months, the nominal estimate is still −0.08. The simulation illustrates the mechanism and does not prove it, because the true model is one in which financial conditions matter by construction. But it shows that if they do matter, a small VAR that omits them can produce the wrong-signed responses found in practice.

High-frequency surprises help identify the shock, but they do not by themselves fix how it is carried forward. Our design therefore does two things. It includes financial conditions in the system, so that they carry the effects of the surprise forward, as in equation (1). And it measures the immediate effect of the surprise on the two-year yield and on financial conditions directly in the announcement window, where policy news is the main force at work. Finally, the measurement of f_t matters. If activity responds to financial conditions relative to r^*, a measure that ignores r^* differs from the true f_t by a slow-moving error, and that error enters equation (1).

2.3 Financial conditions and real activity

A separate literature asks how financial conditions are related to economic activity. Financial conditions predict activity, though unstably: asset prices help forecast output and inflation, but which ones help changes over time (Stock and Watson, 2003). Credit spreads, and especially their excess bond premium component, are strong predictors of activity (Gilchrist and Zakrajšek, 2012). Broad indexes summarize many such series and are used to assess the outlook (Hatzius et al., 2010; Brave and Butters, 2011). Financial conditions are also linked to the risks around growth, not only to its average (Adrian et al., 2019), and their relationship with unemployment may be nonlinear (Kiley, 2020a). Federal Reserve staff now publish a measure of the effect of financial conditions on growth, with weights from large macroeconomic models (Ajello et al., 2023).

This work has a limit that the event-study literature does not. A correlation between financial conditions and later activity need not reflect a causal effect, because financial conditions also respond to the economy. Hatzius et al. (2010) address part of this problem by removing the effects of past growth and inflation from their indicators. The companion note uses the timing in a VAR for the same purpose, and finds that its index helps predict unemployment within the sample, but that out-of-sample gains are modest (FCI note, 2026).

The two literatures have complementary strengths. Work on high-frequency surprises identifies policy shocks cleanly but carries them through a small set of variables. Work on financial conditions studies a broad set of variables but does not isolate policy shocks. We bring them together. The weights in our index come from how its components respond to policy news, and the surprise is carried forward through the estimated relationship between that index and the economy.

3 Empirical approach

The model has three pieces. A surprise moves the two-year yield and a financial conditions index (FCI) by amounts measured in the announcement window: the vector b in equation (14). A daily VAR, equation (15), carries those two series forward. Their monthly averages then enter a monthly VAR for unemployment and inflation, equation (16). Section 3.1 defines the FCI, Section 3.2 sets out the timing assumptions at each frequency, and Section 3.3 gives the model and how we use it. Appendix B lists the data.

3.1 Measuring financial conditions

The FCI is the market index of FCI note (2026): a weighted sum of the 10-year Treasury yield, stock valuations and the real broad dollar. Indexes of this kind are weighted sums of financial variables (Hatzius et al., 2010). They differ mainly in where the weights come from. FCI-G takes them from large macroeconomic models (Ajello et al., 2023). Ours come from the event-study regressions long used to measure how asset prices respond to policy (Kuttner, 2001; Bernanke and Kuttner, 2005; Hanson and Stein, 2015; Nakamura and Steinsson, 2018).

Policy news. Let ΔX_a be the 5×1 vector of changes in the current- and next-meeting federal funds futures and the second to fourth eurodollar futures in the window around announcement a. Following Nakamura and Steinsson (2018), let p_a be the first principal component of these changes, signed so that it moves with the fourth eurodollar future. Policy news is p_a rescaled so that one unit moves the two-year yield y^2 by 1 percentage point:

$$s_a = \gamma\, p_a, \qquad \gamma = \frac{\sum_a p_a\, \Delta y_a^2}{\sum_a p_a^2}. \tag{4}$$

Weights. Let ΔA_a be the vector of changes in the 10-year yield, the S&P 500 and the DXY dollar index in the announcement window. The companion note treats these changes as

$$\Delta A_a = \psi\, s_a + \eta_a, \tag{5}$$

where ψ holds the policy responses and η_a is other news, with covariance matrix Σ. It estimates ψ by regressing each element of ΔA_a on s_a and a constant, and Σ from the residuals, over 253 announcements from January 1996 to September 2026. The weights are those that recover the policy component most precisely (one-factor generalized least squares):

$$w = \frac{\Sigma^{-1}\psi}{\psi'\Sigma^{-1}\psi}. \tag{6}$$

Because $w'\psi = 1$ and s_a is scaled to the two-year yield, a policy move that raises the two-year yield 1 percentage point raises the index by 1 unit. The weights are 0.61 on the 10-year yield, -0.051 per percent on stock valuations and 0.065 per percent on the dollar. Each of the three accounts for about a third of the index's response to policy.

Three versions. In month t, the index in version v is

$$F_t^v = w_B\, B_t^v + w_Q\, Q_t^v + w_D\, D_t, \tag{7}$$

measured as a deviation from its sample mean. Here D_t is 100 times the log of the real broad dollar in every version. With y_t^{10} the 10-year yield, π_t^{10} 10-year expected inflation, and $(E/P)_t$ the inverse of the cyclically adjusted price-earnings ratio (CAPE), the other two terms are:

$$\text{Nominal:} \quad B_t = y_t^{10}, \qquad Q_t = 100 \ln \text{CAPE}_t, \tag{8}$$

$$\text{Real:} \quad B_t = y_t^{10} - \pi_t^{10}, \qquad Q_t = 100 \ln \text{CAPE}_t, \tag{9}$$

$$r^*\text{-gap (baseline):} \quad B_t = y_t^{10} - \pi_t^{10} - r_t^*, \qquad Q_t = -\frac{100}{\overline{E/P}} \left[(E/P)_t - r_t^*\right]. \tag{10}$$

In the r^*-gap version, stocks are measured by the gap between their earnings yield and r^*. The term $100/\overline{E/P}$, with $\overline{E/P}$ the sample average, converts this gap to the units of the log CAPE, so that the weight on stocks is the same in every version. CAPE is a real valuation measure, so stocks do not change between the nominal and real versions. For r^* we use the New York Fed's current Laubach–Williams estimates (Holston et al., 2023). These "one-sided" estimates use only data up to each date, with the model's parameters estimated on the full sample.

The two-year yield. The two-year yield is adjusted in the same steps. With y_t^2 the nominal yield and π_t^2 two-year expected inflation (from the Cleveland Fed; Haubrich et al., 2012),

$$R_t^{\text{nominal}} = y_t^2, \qquad R_t^{\text{real}} = y_t^2 - \pi_t^2, \qquad R_t^{r^*\text{-gap}} = y_t^2 - \pi_t^2 - r_t^*. \tag{11}$$

From monthly to daily. Expected inflation and r^* are not available daily, and they move slowly. We take the monthly gap between R_t^v and y_t^2, interpolate it linearly between the middles of adjacent months, and add it to the daily yield. The daily index is

$$F_\tau^v = w_B\, y_\tau^{10} + w_Q\, 100 \ln \text{CAPE}_\tau + w_D\, 100 \ln \text{DOL}_\tau + o_\tau^v, \tag{12}$$

where DOL_τ is the daily nominal broad dollar. The offset o_τ^v is the monthly difference between the index in equation (7) and the average of the first three terms, interpolated in the same way. It also absorbs the difference between the real and the nominal dollar. The monthly averages of the daily series are very close to the monthly measures: the root-mean-square difference is at most 0.02 unit for the FCI and 0.04 percentage point for the two-year yield. The offsets change little from day to day, so day-to-day moves in the index come from market prices. Table 2 shows the measures used at each frequency.

Table 2: Measures of stocks and the dollar at each frequency

Frequency	Stocks	Dollar
Window	S&P 500	DXY index
Daily	CAPE	Nominal broad dollar, plus offset
Monthly	CAPE, or earnings-yield gap to r^*	Real broad dollar

3.2 Timing assumptions and identification

The shock is the Nakamura–Steinsson policy-news surprise, s_a in equation (4), from the San Francisco Fed's U.S. Monetary Policy Event-Study Database (Acosta et al., 2025). We use the 252

FOMC announcements from January 1996 to July 2026 with all the window measures available. As in the companion note, we leave out unscheduled announcements with no change in the policy rate, and March–April 2020. The one later announcement in the database, September 2026, is left out because too few later days are available. The window runs from 10 minutes before the statement to 60 minutes after the press conference starts. When there is no press conference, it is the 30-minute window around the statement.

Identification comes from timing, at three frequencies. Table 3 summarizes the assumptions.

Table 3: Timing assumptions

Frequency	What can move	Assumption	Equation
Window (minutes)	Two-year yield and FCI, by the amounts measured in the window	Policy news is the only systematic force in the window. Unemployment and inflation do not move.	(14)
Daily	Two-year yield and FCI, each day after the announcement	They depend on their own recent history and on the unemployment and inflation of this month and last month.	(15)
Monthly	Unemployment and inflation, from the month after the announcement	Within a month, unemployment and inflation are set first. They respond to financial variables only with a lag of at least one month.	(16)

The window. The window move in the FCI is computed with the weights in equation (6):

$$\Delta F_a = w_B\, \Delta y_a^{10} + w_Q\, \Delta \mathrm{SP}_a + w_D\, \Delta \mathrm{DXY}_a. \tag{13}$$

These are the measures available within the day. They are also the ones used to estimate the weights, so the window move and the weights are consistent. Percent changes in the S&P 500 equal percent changes in CAPE within a window, because earnings are fixed. The DXY index covers fewer currencies than the broad dollar used in the daily and monthly index, which we treat as a measurement difference.

We express all effects per percentage point on the two-year yield in the window. The effect on the FCI is the ratio of two regression slopes on the surprise, which is an instrumental-variables estimate with s_a as the instrument:

$$b = \begin{pmatrix} 1 \\ b_F \end{pmatrix}, \qquad b_F = \frac{\sum_a (\Delta F_a - \overline{\Delta F})\,(s_a - \bar{s})}{\sum_a (\Delta y_a^2 - \overline{\Delta y^2})\,(s_a - \bar{s})} = 0.99. \tag{14}$$

This is close to one because the weights are normalized that way. It is not exactly one because this regression includes a constant, which the scaling in equation (4) does not, and because it uses one fewer announcement. Expected inflation and r^* move slowly, so we take the window move to be the same in every version.

The day. On the announcement day, the two-year yield and the FCI shift by the vector b. We use the window move rather than the change over the whole day, because the full-day change contains other news. On later days, the two series move as equation (15) implies. The two series affect each other only through the daily lags, so we need no ordering between them within the day. The window fixes their joint move.

The month. Within each month, unemployment and inflation are set first, and markets then respond to them. So in the month of the announcement, unemployment and inflation have not moved. They respond from the next month on, through the monthly average of the daily financial

series. The average depends on when in the month the announcement falls. An announcement late in a month affects few days of that month's average, and an announcement early in the month affects most of them. This is how the model takes the timing of announcements into account, as Jacobson et al. (2026) recommend. The ordering is the familiar recursive one, in which slow-moving macroeconomic variables come before financial variables within the period (Bernanke and Blinder, 1992; Christiano et al., 1999). It also means that the daily equations include the current month's unemployment and inflation, even though those figures are published later. We treat this as a modeling convention: the monthly values summarize economic conditions that markets observe as the month unfolds.

The key assumptions. Two assumptions carry the results. The first is the usual one in a VAR. Once the shock has been identified, the economy responds to a change in the financial variables in the same way whatever caused it. A policy shock identified by a Cholesky ordering or an external instrument is traced through the same estimated dynamics as every other shock, and we do the same. It has a familiar cost. Stock prices partly anticipate future activity, so a move that policy news pushes through stocks need not act on the economy like a move with another cause. This is the same problem as the information effect in any VAR with a high-frequency shock, and as in the FCI note we acknowledge it and set it aside.

The second assumption is specific to this note and has two parts. A standard VAR estimates the effect of the shock on every variable in the system. Here, the surprise reaches unemployment and prices only through the two-year yield and the FCI, and has no direct effect on them in the announcement window. Anything else it moves, such as credit spreads or expectations, is left out. This is easier to accept when the financial variables are broad, but it is a restriction. In addition, the monthly equations see the FCI as a single index. Its weights show how the 10-year yield, stock valuations and the dollar each respond to policy news. They do not show how the economy responds to each of them. So we assume that the three affect unemployment and prices in proportion to those weights. We do not split the FCI into its parts, or into policy-driven and other movements, and we do not add credit variables to the baseline. Appendix C relaxes both parts in turn.

3.3 A mixed-frequency VAR

Structure. The model has a daily block and a monthly block. Let $x_\tau = (R_\tau, F_\tau)'$ hold the two-year yield and the FCI on day τ, in the same version. Let $m_t = (U_t, \pi_t)'$ hold the two monthly variables. The unemployment gap U_t is the unemployment rate minus the CBO estimate of its natural rate. The CBO estimate does not respond to the surprise, so the response of the gap equals the response of the unemployment rate. Inflation is $\pi_t = 1200\,\Delta \ln P_t$, where P_t is the price index for personal consumption expenditures excluding food and energy (core PCE).

The daily block is a VAR in the style of the heterogeneous autoregression of Corsi (2009):

$$x_\tau = \alpha + \sum_{k\in\{1,5,22,66\}} \Phi_k\, \bar{x}^{(k)}_{\tau-1} + \Gamma_0\, m_{t(\tau)} + \Gamma_1\, m_{t(\tau)-1} + \xi_\tau, \tag{15}$$

where $\bar{x}^{(k)}_{\tau-1}$ is the average of x over the k days through $\tau - 1$, and $t(\tau)$ is the month of day τ. Averages over about a day, a week, a month and a quarter capture persistence with few coefficients. Each equation has 13 coefficients with the FCI. When the FCI is left out, x_τ is the two-year yield alone. The calendar has every weekday; missing values, such as on holidays, are filled from the previous observation for up to three days.

The monthly block is

$$m_t = c + \sum_{l=1}^{L} A_l\, m_{t-l} + \sum_{l=1}^{L} G_l\, \bar{x}_{t-l} + \theta_1\, t + \theta_2\, t^2 + \zeta_t, \qquad \bar{x}_t = \frac{1}{n_t} \sum_{\tau \in t} x_\tau, \tag{16}$$

with n_t the number of weekdays in month t. Each equation has $3 + 4L$ coefficients with the FCI (a constant, the two trend terms, and $2L$ each for the macroeconomic and financial lags). The model is a mixed-frequency VAR (Ghysels, 2016). In terms of the mixed-data sampling literature, the weights are equal across days within a month and unrestricted across months (Ghysels et al., 2007; Foroni et al., 2015). The equation contains no current-month term in $\bar{x}_t$, which is the timing assumption of Table 3.

Relation to a monthly VAR. Suppose the daily block were replaced by a monthly equation for $\bar{x}_t$, and every announcement occurred on the first day of the month. With $Z_t = (m_t', \bar{x}_t')'$, the model would then be a standard monthly VAR with a recursive ordering, the macroeconomic variables first, and impact vector $(0, 0, b')'$. The daily block adds two things to that benchmark: the position of each announcement in its month, and fast dynamics of the financial variables, estimated on many more observations. Section 5 estimates this benchmark directly.

This benchmark connects our design to the proxy VAR (Gertler and Karadi, 2015; Mertens and Ravn, 2013; Stock and Watson, 2018). A proxy VAR estimates $Z_t = \sum_l C_l Z_{t-l} + \iota_t$, where ι_t are the monthly residuals, and uses as an instrument z_t the surprises summed over the month. The effect on the FCI per unit effect on the two-year yield is then

$$\frac{\mathrm{Cov}(\iota_{F,t}, z_t)}{\mathrm{Cov}(\iota_{R,t}, z_t)}. \tag{17}$$

The window ratio b_F in equation (14) has the same form, with three differences. It replaces monthly VAR residuals with changes measured inside the window, which do not depend on how strongly the surprise predicts monthly residuals. It sets the entries for unemployment and inflation to zero rather than estimating them. And it dates each shock to its day rather than summing shocks over a month.

The time trend. Every monthly equation includes a smooth quadratic time trend. Over 1994–2026, unemployment, inflation and interest rates contain slow movements that a short monthly model cannot capture. Rates drifted down along with r^*. Inflation was low in the 2010s and rose after 2020. The natural rate of unemployment may have shifted in ways the CBO estimate misses. Without a trend, these slow movements show up in the estimated effect of the financial variables, and trend gets mixed with transmission. A quadratic is the simplest smooth trend that allows inflation to fall and then rise. Because the trend is an assumption, Section 5 reports the main steps without it.

Estimation. We estimate each block by ordinary least squares. The daily block uses January 1990 to September 2026, including 2020. The monthly block uses February 1994 to September 2026, leaving out March–December 2020, which leaves 382 months. We use $L = 3$, 6 or 12 monthly lags.

Tracing the effect of a surprise. The response is computed recursively, as a deviation from a path with no surprise. Let the announcement occur on weekday d_0 of the announcement month, and let $h = 0, 1, \ldots, 36$ count months from then. Tildes denote responses.

1. In each month, unemployment and inflation come first: $\tilde{m}_0 = 0$, and for $h \geq 1$,

$$\tilde{m}_h = \sum_{l=1}^{L} A_l\, \tilde{m}_{h-l} + \sum_{l=1}^{L} G_l\, \tilde{x}^M_{h-l}, \tag{18}$$

with terms dated before the announcement month equal to zero, and $\tilde{x}_h^M$ the response of the monthly average of x.

2. The days of the month follow. Before day d_0, $\tilde{x}_\tau = 0$. On day d_0, $\tilde{x}_\tau = b$ from equation (14). On every later day, with $\tilde{x}_{\tau-1}^{(k)}$ the average of the responses over the k days through $\tau - 1$,

$$\tilde{x}_\tau = \sum_{k \in \{1,5,22,66\}} \Phi_k \, \tilde{x}_{\tau-1}^{(k)} + \Gamma_0 \, \tilde{m}_{h(\tau)} + \Gamma_1 \, \tilde{m}_{h(\tau)-1}, \tag{19}$$

where $h(\tau)$ is the month of day τ, counted from the announcement month.

3. The monthly average $\tilde{x}_h^M$ then enters equation (18) for later months.

Months after the announcement month have 21 weekdays. To avoid computing a path for every date, we form six groups of announcement dates by the weekday of the month, in blocks of four days, and compute the path at each group's average day and month length. With ω_g the share of announcements in group g, the reported response is $\sum_g \omega_g \, \tilde{m}_h^{(g)}$. The response of the core price level after h months, in percent, is $\sum_{j=0}^{h} \tilde{\pi}_j / 12$. Because the model is linear, these responses do not depend on the constant or the trend.

Relation to other methods. In large samples, a VAR of this kind estimates the same effects as local projections, which regress each outcome directly on the shock at each future month (Plagborg-Møller and Wolf, 2021). Section 5 tests the equal weighting of days and reports a direct regression of unemployment on the daily surprises.

Uncertainty. To measure uncertainty, we use a bootstrap with 200 draws, with stars denoting a draw. In each draw, we resample the 252 announcements with replacement, which gives new values b^* in equation (14) and new group shares ω_g^*. We also resample the residuals $\hat{\zeta}_t$ of equation (16) jointly across equations, add them to the fitted values, holding the regressors fixed, and re-estimate the monthly block. The daily block is held at its estimate. We then compute the response for each draw and report percentile intervals. Section 5 reports a fuller version that also re-estimates the daily model and rebuilds the monthly data from the model. The FCI weights and r^* are held fixed throughout. We report 90 percent intervals, as recent studies in this area do (Bauer and Swanson, 2023a; Caldara and Herbst, 2019; Miranda-Agrippino and Ricco, 2021; Ramey, 2016). We also report 68 percent intervals, which Jarociński and Karadi (2020) report alongside 90 percent ones.

4 Results

We report four steps: the nominal two-year yield alone; adding the nominal FCI; real measures of both; and measures of both relative to r^*. Table 4 shows how the unemployment gap and the core PCE price level respond after 12, 24 and 36 months, for 3, 6 and 12 monthly lags, at each step. Figure 1 shows the four key cases.

Table 4: Responses to a policy move that raises the two-year yield 1 percentage point

		12 months			24 months			36 months		
Step	FCI	3	6	12	3	6	12	3	6	12
A. Unemployment gap (percentage points)										
Nominal	none	0.07	−0.13	−0.27**	0.36†	0.10	−0.19	0.53**	0.29	−0.00
	included	0.44**	0.36**	0.28**	0.76**	0.75**	0.63**	0.76**	0.83**	0.77**
Real	none	0.14†	−0.04	−0.20**	0.33**	0.12	−0.13	0.42**	0.23	−0.01
	included	0.44**	0.38**	0.29**	0.65**	0.65**	0.59**	0.57**	0.64**	0.64**
r^*-gap	none	0.19**	0.10	−0.03	0.34**	0.21	0.04	0.38**	0.25†	0.07
	included	0.40**	0.37**	0.36**	0.47**	0.48**	0.50**	0.35**	0.40**	0.43**
B. Core PCE price level (percent)										
Nominal	none	−0.03	−0.01	−0.02	−0.16	−0.16	−0.07	−0.26†	−0.34	−0.18
	included	−0.09	−0.11	−0.11	−0.24	−0.36†	−0.31†	−0.33	−0.54†	−0.49†
Real	none	−0.14†	−0.08	−0.10	−0.25†	−0.23	−0.16	−0.33†	−0.37†	−0.26
	included	−0.17†	−0.17†	−0.18†	−0.30†	−0.40†	−0.36†	−0.35†	−0.54†	−0.49†
r^*-gap	none	−0.27**	−0.22**	−0.23**	−0.45**	−0.51**	−0.52**	−0.55**	−0.70**	−0.80**
	included	−0.34**	−0.33**	−0.33**	−0.53**	−0.68**	−0.72**	−0.57**	−0.85**	−0.97**

Columns: months after the policy move; within each, the number of monthly lags in the model (3, 6 or 12). FCI "included": the FCI of the same step. **: 90 percent interval excludes zero; †: 68 percent interval excludes zero (90 percent does not). Steps are described in Section 3. All versions include a quadratic time trend. Bauer and Swanson (2023a), benchmark (read from their Figure 8): unemployment +0.19, +0.17, +0.15 at 12, 24, 36 months (their 90 percent band excludes zero only at 12 months); CPI (not core PCE) −0.36, −0.48, −0.6 (their bands exclude zero).

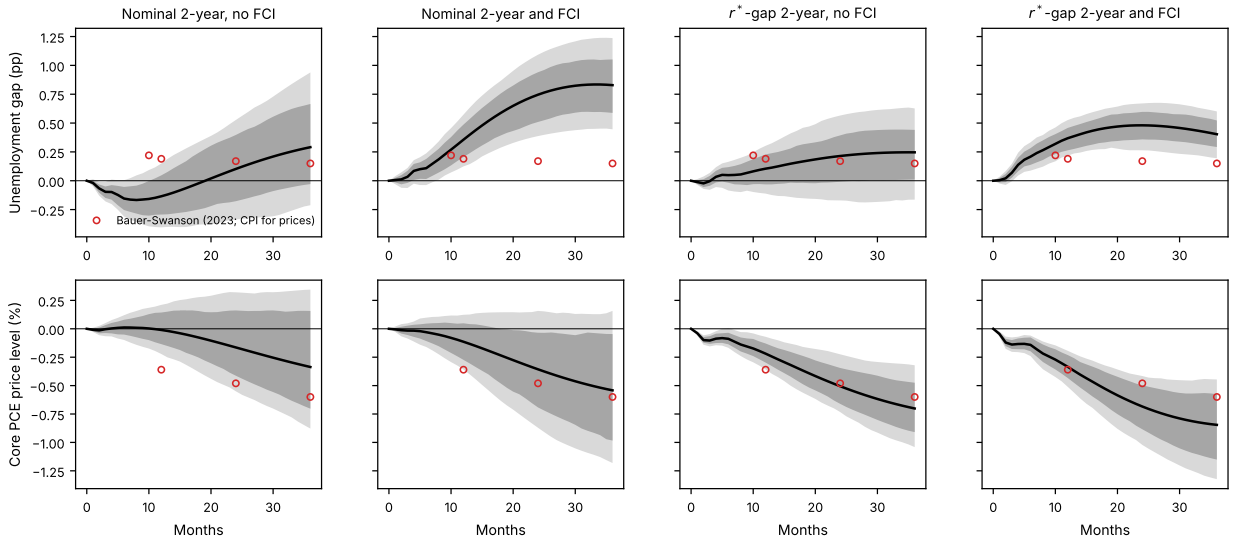


Figure 1: Responses to a policy move that raises the two-year yield 1 percentage point (6 lags). Dark band: 68 percent interval. Light band: 90 percent interval. Circles: Bauer and Swanson (2023a), read from their Figure 8 (unemployment rate; CPI rather than core PCE).

Step 1: the standard approach reproduces the puzzles. Here the nominal two-year yield is the only channel. This is the policy measure that Bauer and Swanson (2023a) use. The unemployment response depends on the number of lags. With 12 lags, it is significantly negative (−0.27 after 12 months), so unemployment falls after a tightening. The price level barely moves.

This is the post-1994 pattern documented by Ramey (2016), and the one Bauer and Swanson (2023a) find with their unadjusted surprises. Industrial production rises significantly (Appendix Table A1).

Step 2: add financial conditions. Adding the FCI makes the unemployment response standard, large and significant at the 90 percent level at 12, 24 and 36 months and for every number of lags. Unemployment rises 0.28 to 0.44 percentage point after 12 months and 0.6 to 0.8 after 24 to 36 months. The price level falls 0.2 to 0.5 percent after 24 to 36 months. This fall is significant only at the 68 percent level, and only with 6 and 12 lags. Industrial production does not fall.

Step 3: real rates. Measuring the two-year yield and the FCI net of expected inflation changes little. The companion note finds the same: after 1990, its nominal and real indexes are close. The price fall becomes significant at the 68 percent level at 12, 24 and 36 months and for every number of lags.

Step 4: rates relative to r^*. Next we measure the two-year yield and the FCI relative to r^*. Unemployment rises 0.36 to 0.40 percentage point after 12 months, 0.47 to 0.50 after 24 months and 0.35 to 0.43 after 36 months. These numbers are nearly the same for every lag length. The core price level falls 0.33 to 0.34 percent after 12 months, 0.53 to 0.72 percent after 24 months and 0.57 to 0.97 percent after 36 months. All of these are significant at the 90 percent level for every lag length. The unemployment response is smaller and fades sooner than with the nominal FCI. Without the FCI, the r^*-gap two-year yield gives a strong price effect. But the unemployment effect is then weak and depends on the number of lags (it is significant only with 3 lags).

Comparison with Bauer and Swanson (2023a). Their benchmark model covers 1973–2020. It uses surprises from FOMC announcements and Chair speeches, with the predictable part removed. Per percentage point on the two-year yield, it gives an unemployment response of about 0.19 percentage point after 12 months, 0.17 after 24 months and 0.15 after 36 months. It gives a CPI decline of about 0.36, 0.48 and 0.6 percent. Our step-4 price effects (for core PCE) are similar after 12 and 24 months and larger after 36 months (-0.57 to -0.97). Our unemployment effect is about twice as large after 12 months and 2.5 to 3 times as large after 24 to 36 months. Our intervals exclude zero through 36 months. Theirs include zero after about a year and a half. Their estimates use surprises from 1988 to 2019, Chair speeches and removal of the predictable part. Ours use FOMC announcement windows alone, with no such adjustment.

5 Robustness

How should the two-year yield be adjusted? Table 5 and Table 6 keep the FCI as one of the companion note's three indexes. The r^*-gap FCI always uses the Laubach–Williams r^*. The tables change only how the two-year yield is adjusted:

- no adjustment (nominal);
- minus expected inflation and the Laubach–Williams r^* (baseline);
- minus expected inflation and Kiley (2020b)'s estimate of r^*;
- minus a market-based measure of the neutral rate: the one-year rate nine years ahead (the 9–10-year forward rate), from the Federal Reserve Board's estimated yield curve (Gürkaynak et al., 2007). This rate is nominal, so it replaces both the expected-inflation and r^* adjustments;
- minus the average of the two-year yield over the past 8 years.

Three results stand out.

First, financial conditions are essential for unemployment. Without the FCI, the unemployment response is weak with a model-based r^*. It is significantly negative with the nominal rate, the forward rate or the 8-year average. With any of the FCIs, it is positive at every horizon and significant at the 90 percent level in almost all cases.

Second, the price effect is strong whenever the FCI is measured relative to r^*. With the r^*-gap FCI, the price level falls 0.3 to 0.7 percent after 24 months. This holds for every adjustment of the two-year yield, and even with no adjustment. This fall is significant at the 90 percent level for every lag length. With the nominal or real FCI, the price effect is strong when the two-year yield is adjusted with a model-based r^*. It is weaker with the forward rate or the 8-year average. The forward rate also includes a term premium (the extra yield investors require to hold long-term bonds), which fell over the sample.

Third, Kiley's r^* gives results close to those with Laubach–Williams, though somewhat weaker for prices. The r^*-gap FCI always uses the Laubach–Williams r^*. So these results show that how the two-year yield is adjusted matters less once the FCI is measured relative to r^*. They do not show that the choice of r^* in the FCI is unimportant.

Table 5: Different adjustments of the two-year yield: unemployment gap (percentage points)

	12 months			24 months			36 months		
FCI	3	6	12	3	6	12	3	6	12
Two-year yield: nominal									
none	0.07	−0.13	−0.27**	0.36†	0.10	−0.19	0.53**	0.29	−0.00
nominal	0.44**	0.36**	0.28**	0.76**	0.75**	0.63**	0.76**	0.83**	0.77**
real	0.41**	0.32**	0.25†	0.69**	0.67**	0.57**	0.68**	0.74**	0.70**
r^*-gap	0.39**	0.31**	0.29**	0.58**	0.57**	0.55**	0.52**	0.57**	0.60**
Two-year yield: Laubach–Williams r^*-gap (baseline)									
none	0.19**	0.10	−0.03	0.34**	0.21	0.04	0.38**	0.25†	0.07
nominal	0.48**	0.43**	0.35**	0.63**	0.63**	0.58**	0.49**	0.53**	0.53**
real	0.46**	0.42**	0.36**	0.59**	0.59**	0.57**	0.45**	0.49**	0.51**
r^*-gap	0.40**	0.37**	0.36**	0.47**	0.48**	0.50**	0.35**	0.40**	0.43**
Two-year yield: Kiley r^*-gap									
none	0.13†	−0.02	−0.15†	0.26†	0.07	−0.11	0.30**	0.13	−0.03
nominal	0.44**	0.38**	0.32**	0.58**	0.60**	0.58**	0.43**	0.50**	0.56**
real	0.44**	0.38**	0.32**	0.55**	0.58**	0.57**	0.40**	0.48**	0.54**
r^*-gap	0.39**	0.36**	0.33**	0.45**	0.48**	0.52**	0.31**	0.39**	0.45**
Two-year yield: minus 9–10-year forward rate									
none	−0.20**	−0.20†	−0.23**	−0.25**	−0.17	−0.23†	−0.24**	−0.11	−0.14
nominal	0.26**	0.25**	0.18†	0.33**	0.42**	0.35**	0.26**	0.35**	0.35**
real	0.26**	0.26**	0.20†	0.31**	0.40**	0.35†	0.24**	0.33**	0.34†
r^*-gap	0.22**	0.23**	0.26**	0.26**	0.35**	0.40**	0.21**	0.30**	0.37**
Two-year yield: minus average over past 8 years									
none	−0.16†	−0.28**	−0.30**	−0.05	−0.18	−0.30†	0.02	−0.06	−0.18
nominal	0.30**	0.25**	0.21**	0.39**	0.43**	0.40**	0.27**	0.35**	0.38**
real	0.27**	0.21**	0.17†	0.36**	0.39**	0.36**	0.26**	0.33**	0.36**
r^*-gap	0.22**	0.17**	0.21**	0.33**	0.36**	0.41**	0.29**	0.37**	0.46**

See Table 4. The FCI is always one of the companion note's three indexes; the r^*-gap FCI uses the Laubach–Williams r^*. All specifications include the quadratic trend.

Table 6: Different adjustments of the two-year yield: core PCE price level (percent)

	12 months			24 months			36 months		
FCI	3	6	12	3	6	12	3	6	12
Two-year yield: nominal									
none	−0.03	−0.01	−0.02	−0.16	−0.16	−0.07	−0.26†	−0.34	−0.18
nominal	−0.09	−0.11	−0.11	−0.24	−0.36†	−0.31†	−0.33	−0.54†	−0.49†
real	−0.09	−0.11	−0.13	−0.23	−0.34†	−0.33	−0.31	−0.51†	−0.50
r^*-gap	−0.20**	−0.22**	−0.21**	−0.32**	−0.50**	−0.52**	−0.32†	−0.61†	−0.72**
Two-year yield: Laubach–Williams r^*-gap (baseline)									
none	−0.27**	−0.22**	−0.23**	−0.45**	−0.51**	−0.52**	−0.55**	−0.70**	−0.80**
nominal	−0.25**	−0.28**	−0.30**	−0.42**	−0.59**	−0.66**	−0.50**	−0.75**	−0.92**
real	−0.26**	−0.29**	−0.33**	−0.44**	−0.59**	−0.70**	−0.52**	−0.75**	−0.97**
r^*-gap	−0.34**	−0.33**	−0.33**	−0.53**	−0.68**	−0.72**	−0.57**	−0.85**	−0.97**
Two-year yield: Kiley r^*-gap									
none	−0.23**	−0.16†	−0.18**	−0.37**	−0.34†	−0.29†	−0.44**	−0.48†	−0.41†
nominal	−0.23**	−0.24**	−0.21†	−0.39**	−0.50**	−0.41†	−0.43**	−0.61**	−0.53†
real	−0.25**	−0.25**	−0.25**	−0.40**	−0.51**	−0.46†	−0.44**	−0.62**	−0.58†
r^*-gap	−0.34**	−0.33**	−0.29**	−0.48**	−0.62**	−0.58**	−0.47**	−0.69**	−0.68**
Two-year yield: minus 9–10-year forward rate									
none	0.01	−0.04	−0.08	0.04	−0.04	−0.06	0.05	−0.06	−0.07
nominal	−0.05	−0.12	−0.14†	−0.07	−0.21	−0.25	−0.06	−0.22	−0.31
real	−0.06	−0.14†	−0.18†	−0.07	−0.21	−0.30†	−0.06	−0.21	−0.37
r^*-gap	−0.22**	−0.27**	−0.30**	−0.29**	−0.48**	−0.62**	−0.30†	−0.53**	−0.81**
Two-year yield: minus average over past 8 years									
none	−0.08	−0.06	−0.08	−0.25†	−0.27†	−0.23†	−0.37†	−0.47†	−0.45†
nominal	−0.13†	−0.15†	−0.16†	−0.27†	−0.38**	−0.39†	−0.31†	−0.49**	−0.56†
real	−0.14†	−0.15†	−0.18†	−0.28†	−0.37**	−0.42†	−0.32†	−0.49**	−0.61†
r^*-gap	−0.25**	−0.24**	−0.24**	−0.46**	−0.61**	−0.63**	−0.56**	−0.83**	−0.96**

See Table 5. The 8-year average and the Laubach–Williams trend in the two-year yield (two-year expected inflation plus r^*) are close until 2015 (averages 5.3 and 5.5 percent over 1994–2007; 2.3 and 2.5 over 2008–2015) and diverge after 2016 (1.4 and 3.6).

The pandemic and recent years. The baseline drops March–December 2020 from the months used to fit the monthly equations. But 2020 values still enter the 2021 equations as lags. Table 7 therefore also drops 2021. It then ends the sample in December 2022, and separately in December 2019. In those two shorter samples, the daily model is also estimated only over the shorter period.

With the r^*-gap two-year yield and FCI, the unemployment response holds up. After 12 months, it is 0.26 to 0.31 percentage point when 2021 is dropped, 0.32 to 0.34 when the sample ends in 2022, and 0.42 to 0.44 when it ends in 2019. It is significant at the 90 percent level at almost every horizon and lag length. Without the FCI, its 90 percent interval includes zero in all three samples.

The price effect is unchanged when the sample ends in 2022, so the data since 2023 play no special role. It stays significant with 6 and 12 lags when 2021 is dropped (−0.43 and −0.63 percent after 24 months). But in the sample that ends in 2019, the price level barely moves. So the significant price effect comes from including 2020–22.

Figure 2 shows why these years are informative. After 2021, the r^*-gap two-year yield and the FCI rose by about 4 percentage points. This was the largest swing in the stance of policy relative to r^* in the sample, and core inflation then fell. Figure 3 drops one calendar year at a time. No single

year drives the results. The unemployment response after 12 months ranges from 0.26 (dropping 2021) to 0.41. The price level after 24 months ranges from −0.43 (dropping 2021) to −0.86. The 90 percent intervals exclude zero in every case. So the price result rests on the 2020–22 episode as a whole—a large tightening relative to r^* followed by falling inflation—not on any one year.

Table 7: Different samples: r^*-gap two-year yield, without and with the r^*-gap FCI

		12 months			24 months			36 months		
Sample	FCI	3	6	12	3	6	12	3	6	12
A. Unemployment gap (percentage points)										
Baseline (excl. Mar–Dec 2020)	none	0.19**	0.10	−0.03	0.34**	0.21	0.04	0.38**	0.25†	0.07
	r^*-gap	0.40**	0.37**	0.36**	0.47**	0.48**	0.50**	0.35**	0.40**	0.43**
Also drop 2021	none	0.03	−0.03	−0.09	0.09	−0.01	−0.09	0.12	−0.01	−0.11
	r^*-gap	0.31**	0.26**	0.30**	0.38**	0.33**	0.38**	0.29**	0.28†	0.30**
End in 2022	none	0.11	0.07	−0.04	0.19†	0.11	−0.01	0.21†	0.13	0.01
	r^*-gap	0.32**	0.33**	0.34**	0.35**	0.38**	0.43**	0.23**	0.30**	0.37**
End in 2019	none	0.05	0.05	0.00	0.09	0.14	0.14	0.11	0.19	0.28
	r^*-gap	0.42**	0.42**	0.44**	0.45**	0.50**	0.60**	0.34**	0.42**	0.59**
B. Core PCE price level (percent)										
Baseline (excl. Mar–Dec 2020)	none	−0.27**	−0.22**	−0.23**	−0.45**	−0.51**	−0.52**	−0.55**	−0.70**	−0.80**
	r^*-gap	−0.34**	−0.33**	−0.33**	−0.53**	−0.68**	−0.72**	−0.57**	−0.85**	−0.97**
Also drop 2021	none	−0.11†	−0.09	−0.13†	−0.18†	−0.25†	−0.35**	−0.22†	−0.38†	−0.59**
	r^*-gap	−0.16†	−0.21**	−0.28**	−0.24†	−0.43**	−0.63**	−0.29†	−0.60**	−0.93**
End in 2022	none	−0.25**	−0.20**	−0.21**	−0.42**	−0.49**	−0.51**	−0.51**	−0.70**	−0.83**
	r^*-gap	−0.34**	−0.33**	−0.30**	−0.51**	−0.68**	−0.71**	−0.56**	−0.87**	−1.02**
End in 2019	none	0.02	0.02	0.08†	0.06	0.02	0.03	0.10	0.02	−0.06
	r^*-gap	−0.02	−0.03	−0.05	0.02	−0.00	−0.12	0.03	−0.03	−0.22†

See Table 4. In the "End in" samples, the daily model is also estimated over the shorter period.

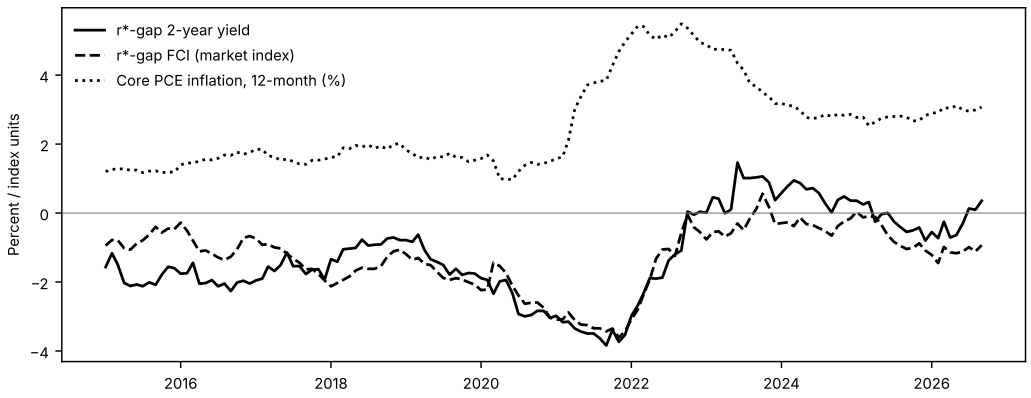


Figure 2: The r^*-gap two-year yield, the r^*-gap FCI and 12-month core PCE inflation, 2015–2026.

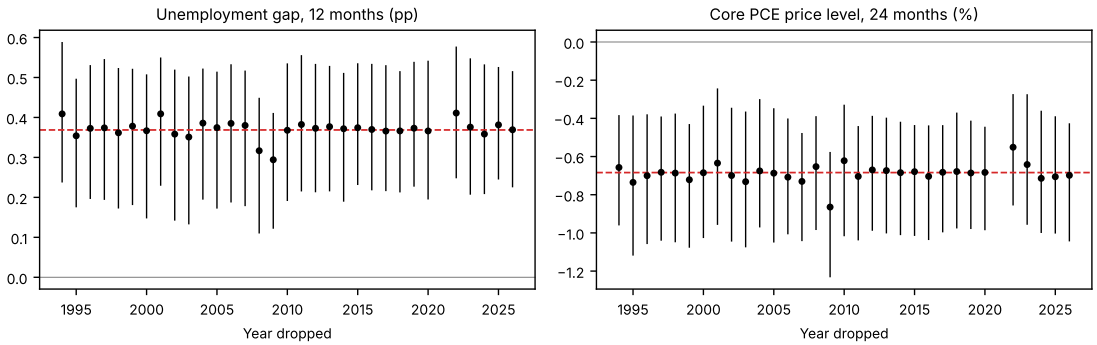


Figure 3: Dropping one year at a time. Baseline version (r^*-gap two-year yield and FCI, 6 lags; 100 bootstrap draws rather than 200, to limit computation). Each point drops one calendar year from the monthly equations. Bars: 90 percent intervals. Dashed line: full-sample estimate.

Industrial production. Appendix Table A1 adds industrial production to the monthly model. Without the FCI, industrial production rises after a tightening when the two-year yield is nominal or real. It does not move significantly when the two-year yield is measured relative to r^*. With the nominal or real FCI, it does not fall significantly. Only at the r^* step does it fall: by 0.84 to 1.05 percent after 12 months, significant at the 90 percent level for every lag length. Bauer and Swanson (2023a) find about 1.5 percent. In our model, industrial production recovers within about three years. In earlier exploratory work, we also added a commodity price index. Commodity prices then rose after a tightening. Our assumptions are least plausible for prices like these, which look ahead and respond strongly to demand.

Other surprise measures. Appendix Table A2 replaces the Nakamura–Steinsson surprise with three others and keeps everything else the same:

- the surprise with the part predicted by the Fed staff's forecasts for each FOMC meeting (the Greenbook) removed;
- the surprise of Bauer and Swanson (2023a), with the part predicted by public data removed (their posted series, which covers 201 of our announcements through 2019);
- the "pure policy" surprise of Jarociński and Karadi (2020), which keeps only announcements where the two-year yield and the S&P 500 move in opposite directions (157 announcements).

The choice of surprise series can matter for estimated effects (Bu et al., 2021; Brennan et al., 2025). In our design, a surprise enters only through its window moves, scaled to a 1 percentage point rise in the two-year yield, and through the timing of announcements. So this check really tests how much the FCI moves per point on the two-year yield. That ratio is 0.99 for the baseline, 1.07 for the Bauer–Swanson surprise and 1.09 for the Jarociński–Karadi surprise. The last ratio is higher partly by design: that surprise keeps announcements where stocks fall when yields rise, and stocks enter the FCI with a negative sign.

With these ratios, the effects are slightly larger than in the baseline. At the r^* step, unemployment rises 0.40 to 0.45 percentage point after 12 months, and the price level falls 0.55 to 0.77 percent after 24 months. The Greenbook-cleaned surprise gives results within 0.02 of the baseline. Without the FCI, all the surprises give almost identical effects. So our results do not depend on how the surprise is cleaned. Cleaning matters here only through the ratio of the FCI move to the two-year-yield move in the window.

A fuller measure of uncertainty. Appendix Table A4 uses 1,000 draws instead of 200.

In each draw, besides resampling the announcements, it re-estimates the daily model on randomly redrawn blocks of 66 consecutive days. It also rebuilds the monthly data from the estimated model, randomly flipping the signs of the errors, and re-estimates the monthly equations. The intervals barely change. At the r^* step, every ** and † marker is the same. Elsewhere, a few markers move between the 68 and 90 percent levels. The FCI weights and r^* are still held fixed.

Relaxing the key assumptions. Appendix C tests the two restrictions specific to this note and adds a monthly credit measure. First, we add the daily gap between the Baa corporate yield and the 10-year Treasury yield, and let it respond on impact to the surprise. The results barely change: at the r^* step, unemployment still rises 0.28 to 0.35 percentage point after 12 months and the price level still falls 0.48 to 0.65 percent after 24 months, significant for all lag lengths. Second, we replace the FCI with its three parts, entered separately. The signs are unchanged, but the estimates are less precise, and the price result is significant at the 90 percent level only with 12 lags. A test of the proportional-weights restriction does not reject it at the 5 percent level for unemployment at any step, or for either variable at the r^* step. Third, adding the monthly excess bond premium as a third monthly variable removes the wrong-signed unemployment response without the FCI, but leaves the results with the FCI almost unchanged.

Longer sample for the monthly equations. The baseline fits the monthly equations on 1994–2026, close to the years of the surprises (1996–2026). Some studies instead estimate the dynamics on a longer sample and apply the surprises only to the recent period. Appendix Tables A5 and A6 do this. The surprises, the announcement windows and the daily model are unchanged. Only the months used to fit the unemployment and inflation equations start earlier, in 1973, 1977 or 1982. Before the daily data exist, the model uses monthly averages instead. Three splices are needed and are described in the table notes. We also run the 1973 start without the time trend.

Longer samples change the picture in four ways. First, the unemployment puzzle depends on the 1994 start. With the nominal two-year yield alone, the 1994 estimates are significantly wrong-signed at 12 months with 12 lags (-0.27), and the sign changes with the lag length. With the 1973 start, no point estimate is negative. But the responses are small ($+0.02$ to $+0.09$ at 12 months, $+0.10$ to $+0.16$ at 24 months) and mostly not significant. Second, the FCI still matters for unemployment in the long sample. The response has the standard sign in every sample, at every horizon and lag length, and it is significant at 12 months with 3, 6 and 12 lags. But the long-sample effects are smaller. At the r^* step the response is 0.11 to 0.27 percentage point after 12 months with a 1973 start, 0.19 to 0.28 with a 1977 start and 0.21 to 0.29 with a 1982 start, against 0.36 to 0.40 in the baseline. At 36 months the 1973 effects are 0.02 to 0.08, against 0.35 to 0.43. Third, a price puzzle appears with the nominal yield in the 1973 sample. The price level rises, by 0.15 percent at 24 months with 3 lags, though not significantly. The baseline shows only a non-significant fall. Adjusting for r^* removes the puzzle in both samples. At the r^* step, with and without the FCI, the price level falls significantly at 24 months with 6 and 12 lags in every long sample. With the FCI the fall is 0.31 to 0.54 percent, against 0.53 to 0.72 in the baseline. In the 1973 and 1977 samples the real step also gives significant price falls, which the baseline does not. Fourth, the signs agree across starts whenever the FCI is included or the yield is adjusted for r^*, and the 1994 effects are larger by a factor of about 1.5 to 4, depending on the horizon. Industrial production falls in the 1973 and 1977 samples when the FCI is included (about 0.4 to 0.9 percent after 12 months). It does not fall significantly in the 1982 sample.

A purely monthly VAR. How much of our result comes from the mixed-frequency model, and how much from measuring the impact in the window? To find out, we estimate a monthly VAR on the same sample, with the same quadratic trend and lags: $Z_t = (m_t', \bar{x}_t')'$ in the notation of Section 3.3, with and without the FCI, at each step. We then impose four sets of restrictions on the immediate effect of the surprise on Z_t, written S. The surprise summed over the month is the

instrument z_t, which is zero in months without an announcement.

- *Standard proxy VAR.* Every element S_i is estimated as in equation (17), with variable i in the numerator.
- *Proxy with timing restrictions.* The same, but the elements for unemployment and inflation are set to zero.
- *Window impact, announcement timing.* $S = \phi\,(0, 0, b')'$, with b from equation (14). Here $\phi = 0.46$ is the average share of the month that follows the announcement, because a surprise moves only part of that month's average.
- *Window impact, first day of the month.* $S = (0, 0, b')'$. This treats every surprise as if it occurred on the first day of the month, and is an upper bracket.

For the proxy cases, we report 90 percent Anderson–Rubin sets, which remain valid when the instrument is weak; they are conditional on the VAR coefficients. For the window cases, we use the bootstrap of Section 3.3. The proxy cases are per percentage point on the monthly average of the two-year yield, as is the first-day case. The announcement-timing case is per percentage point in the window, as in the baseline. Appendix Table A7 gives the results.

Two findings stand out. First, the monthly proxy is weak. The first-stage F statistic is 4.9 to 6.3 without the FCI and 1.7 to 6.4 with it, well below the usual threshold of 10, and it is 5.3 to 8.3 when the funds rate and the current-meeting surprise replace the two-year yield and the policy-news surprise. Many of the sets are wide or unbounded. The proxy also implies a much smaller response of the FCI, 0.13 to 0.62 unit per percentage point on the two-year yield, than the 0.99 measured in the windows. With this estimated impact, adding the FCI does not produce a standard unemployment response. At the r^* step, the 12-month response is between -0.11 and $+0.02$. With the funds rate and the current-meeting surprise, it is negative, per percentage point on the funds rate, and significantly so at the nominal step, though the sets include zero at the r^* step. So adding the FCI to a standard monthly proxy VAR is not enough.

Second, a monthly VAR with the impact measured in the window gives results close to ours. With the r^*-gap two-year yield and FCI, unemployment rises 0.21 to 0.23 percentage point after 12 months with announcement timing, and 0.47 to 0.51 with first-day timing. Our baseline of 0.36 to 0.40 lies between these. The price level falls 0.29 to 0.39 percent after 24 months with announcement timing and 0.62 to 0.84 with first-day timing, again bracketing the baseline of 0.53 to 0.72. All of these are significant at the 90 percent level. Without the FCI, the unemployment response is small or has the wrong sign, whichever timing we use, and it is significant at the 90 percent level at the r^* step only with 3 lags. The result therefore depends on two things together: including the FCI, and measuring its response to the surprise in the window. The daily block appears mainly to refine the timing.

The time trend. Appendix Table A3 shows the main steps without the trend. The nominal and real steps depend much more on the trend than the r^* step does. This suggests that measuring rates relative to r^* does much of the work of the trend.

Daily weights and a direct regression. To test equal weighting of days, we replace $\bar{x}_t$ in equation (16) with a weighted average whose weight on the d-th weekday of a month is proportional to $1 + \delta\,(2d - n_t - 1)/(n_t - 1)$, choose δ on a grid from -1 to 1, and test $\delta = 0$ with an F test. This check uses 12 lags, the nominal two-year yield, the r^*-gap FCI and no trend. The test does not reject equal weights for unemployment ($F = 2.1$). It does reject them for inflation ($F = 4.6$, against a 5 percent critical value of 3.9), but the gain in fit is very small: the R^2 rises from 0.435 to 0.443.

We also tried a method that does not need the assumptions of Section 3.2: a direct regression of monthly unemployment on the daily surprises (Chudik and Georgiadis, 2022; Chudik and Kilian, 2026). With s_a the surprise on day τ_a, it estimates a daily response β_ℓ from

$$U_t = \frac{1}{n_t} \sum_{\tau \in t} \sum_{a:\, \tau_a \le \tau} \beta_{\tau - \tau_a}\, s_a + \text{error}, \tag{20}$$

with β_ℓ piecewise linear in ℓ and knots at 0, 1, 2, 3, 6, 9, 12, 18, 24, 30 and 36 months. It gives wrong-signed estimates, driven largely by the 2007–09 announcements. Its intervals are also more than ten times as wide as ours, so it cannot tell our estimates apart from zero.

Other checks. In earlier versions of the model, we added the gap between Baa corporate and Treasury yields to the daily model without any response on impact, because no intraday credit data exist. This lowered the unemployment response by a third to a half. Appendix C instead lets the spread respond on the announcement day, and also adds the monthly excess bond premium. The results then change little when the FCI is included. Credit is still the most important channel we leave out of the baseline (Gilchrist and Zakrajšek, 2012; Caldara and Herbst, 2019). We also tried replacing the daily model's path for the financial variables in the first one to two months with the path estimated directly from the data. That gives larger but much less precise effects.

6 Conclusion

Interest-rate surprises around Federal Reserve announcements measure policy shocks well. The hard part has been tracing their effects on the economy. We keep the surprise as it is and change only how its effects are traced. Two standard parts of the transmission mechanism explain much of the post-1994 puzzle. Policy works through financial conditions as a whole. And what matters for spending and inflation is the level of rates and financial conditions relative to a slow-moving neutral rate. When we trace surprises through a broad financial conditions index measured relative to r^*, the effects on unemployment, prices and industrial production have the standard signs. They are precisely estimated, the unemployment effects barely depend on the number of lags, and they are similar in size to the estimates of Bauer and Swanson (2023a).

The two parts play different roles. Financial conditions are essential for the unemployment effect, in every version and sample we try. Measuring financial conditions relative to r^* is essential for the price effect. When the FCI is measured relative to the Laubach–Williams r^*, the price effect is strong however the two-year yield is adjusted. The unemployment effect holds up however we treat the pandemic. The price effect depends mainly on the years since 2020, when the stance of policy relative to r^* moved most.

These findings complement the work on information effects and predictable surprises. Several mechanisms can operate at once. A natural next step is to trace cleaned or separated surprises through our approach. Our results have limits. They rest on the two key assumptions: policy reaches the economy through the financial variables in the model, and the economy responds to them the same way whatever moved them. They also rest on the time trend. Their baseline leaves out credit spreads, although adding a daily credit spread or the monthly excess bond premium changes little when the FCI is included (Appendix C). And our intervals do not include uncertainty about r^* or the FCI weights. The broader lesson is that the fall in r^* over the past three decades is more than a fact about the level of interest rates. It also makes the effects of policy harder to measure, unless rates are measured relative to r^*.

References

Acosta, M., A. Ajello, M. Bauer, F. Loria and S. Miranda-Agrippino (2025), "Financial Market Effects of FOMC Communication: Evidence from a New Event-Study Database," Federal Reserve Bank of San Francisco Working Paper 2025-30.

Adrian, T., N. Boyarchenko and D. Giannone (2019), "Vulnerable Growth," American Economic Review 109(4), 1263–1289.

Ajello, A., M. Cavallo, G. Favara, W. Peterman, J. Schindler and N. Sinha (2023), "A New Index to Measure U.S. Financial Conditions," FEDS Notes, Board of Governors of the Federal Reserve System, June 30.

Ando, A. and F. Modigliani (1963), "The 'Life Cycle' Hypothesis of Saving: Aggregate Implications and Tests," American Economic Review 53(1), 55–84.

Auclert, A. (2019), "Monetary Policy and the Redistribution Channel," American Economic Review 109(6), 2333–2367.

Bauer, M. D. and G. D. Rudebusch (2020), "Interest Rates under Falling Stars," American Economic Review 110(5), 1316–1354.

Bauer, M. D. and E. T. Swanson (2023a), "A Reassessment of Monetary Policy Surprises and High-Frequency Identification," NBER Macroeconomics Annual 2022, Vol. 37, 87–155.

Bauer, M. D. and E. T. Swanson (2023b), "An Alternative Explanation for the 'Fed Information Effect'," American Economic Review 113(3), 664–700.

Bernanke, B. S. and K. N. Kuttner (2005), "What Explains the Stock Market's Reaction to Federal Reserve Policy?" Journal of Finance 60(3), 1221–1257.

Bernanke, B. S. and A. S. Blinder (1992), "The Federal Funds Rate and the Channels of Monetary Transmission," American Economic Review 82(4), 901–921.

Bernanke, B. S., J. Boivin and P. Eliasz (2005), "Measuring the Effects of Monetary Policy: A Factor-Augmented Vector Autoregressive (FAVAR) Approach," Quarterly Journal of Economics 120(1), 387–422.

Boivin, J., M. T. Kiley and F. S. Mishkin (2010), "How Has the Monetary Transmission Mechanism Evolved Over Time?" in B. M. Friedman and M. Woodford (eds.), Handbook of Monetary Economics, Vol. 3A, 369–422.

Brave, S. and R. A. Butters (2011), "Monitoring Financial Stability: A Financial Conditions Index Approach," Federal Reserve Bank of Chicago Economic Perspectives 35(1), 22–43.

Brennan, C. M., M. M. Jacobson, C. Matthes and T. B. Walker (2025), "Monetary Policy Shocks: Data or Methods?" B.E. Journal of Macroeconomics 25(2), 595–659.

Bu, C., J. Rogers and W. Wu (2021), "A Unified Measure of Fed Monetary Policy Shocks," Journal of Monetary Economics 118, 331–349.

Caballero, R. J., T. E. Caravello and A. Simsek (2026), "FCI-star," NBER Working Paper 33952 (version of January 12, 2026).

Caldara, D. and E. Herbst (2019), "Monetary Policy, Real Activity, and Credit Spreads: Evidence from Bayesian Proxy SVARs," American Economic Journal: Macroeconomics 11(1), 157–192.

Christiano, L. J., M. Eichenbaum and C. L. Evans (1999), "Monetary Policy Shocks: What Have We Learned and to What End?" in J. B. Taylor and M. Woodford (eds.), Handbook of Macroeconomics, Vol. 1A, 65–148.

Christiano, L. J., M. Eichenbaum and C. L. Evans (2005), "Nominal Rigidities and the Dynamic Effects of a Shock to Monetary Policy," Journal of Political Economy 113(1), 1–45.

Chudik, A. and G. Georgiadis (2022), "Estimation of Impulse Response Functions When Shocks Are Observed at a Higher Frequency Than Outcome Variables," Journal of Business and Economic Statistics 40(3), 965–979.

Chudik, A. and L. Kilian (2026), "Mean Group and Pooled Mixed-Frequency Estimators of Responses of Low-Frequency Variables to High-Frequency Shocks," Federal Reserve Bank of Dallas Working Paper 2603.

Corsi, F. (2009), "A Simple Approximate Long-Memory Model of Realized Volatility," Journal of Financial Econometrics 7(2), 174–196.

Companion note (2026), "A Simple Market-Based Financial Conditions Index: Weights from Policy News, Dynamics from a Transparent VAR, Measured Relative to r^*," draft.

Faust, J., E. T. Swanson and J. H. Wright (2004), "Identifying VARs Based on High Frequency Futures Data," Journal of Monetary Economics 51(6), 1107–1131.

Fernández-Villaverde, J., J. F. Rubio-Ramírez, T. J. Sargent and M. W. Watson (2007), "ABCs (and Ds) of Understanding VARs," American Economic Review 97(3), 1021–1026.

Fleming, J. M. (1962), "Domestic Financial Policies under Fixed and under Floating Exchange Rates," IMF Staff Papers 9(3), 369–380.

Foroni, C., M. Marcellino and C. Schumacher (2015), "Unrestricted Mixed Data Sampling (MIDAS): MIDAS Regressions with Unrestricted Lag Polynomials," Journal of the Royal Statistical Society Series A 178(1), 57–82.

Francis, N., E. Ghysels and M. T. Owyang (2011), "The Low-Frequency Impact of Daily Monetary Policy Shocks," Federal Reserve Bank of St. Louis Working Paper 2011-009.

Friedman, M. (1957), *A Theory of the Consumption Function*, Princeton University Press.

Galí, J. and T. Monacelli (2005), "Monetary Policy and Exchange Rate Volatility in a Small Open Economy," Review of Economic Studies 72(3), 707–734.

Gazzani, A. and A. Vicondoa (2020), "Bridge Proxy-SVAR: Estimating the Macroeconomic Effects of Shocks Identified at High-Frequency," Banca d'Italia Working Paper 1274.

Gertler, M. and P. Karadi (2015), "Monetary Policy Surprises, Credit Costs, and Economic Activity," American Economic Journal: Macroeconomics 7(1), 44–76.

Ghysels, E. (2016), "Macroeconomics and the Reality of Mixed Frequency Data," Journal of Econometrics 193(2), 294–314.

Ghysels, E., A. Sinko and R. Valkanov (2007), "MIDAS Regressions: Further Results and New Directions," Econometric Reviews 26(1), 53–90.

Gilchrist, S. and E. Zakrajšek (2012), "Credit Spreads and Business Cycle Fluctuations," American Economic Review 102(4), 1692–1720.

Gürkaynak, R. S., B. Sack and E. T. Swanson (2005), "Do Actions Speak Louder Than Words? The Response of Asset Prices to Monetary Policy Actions and Statements," International Journal of Central Banking 1(1), 55–93.

Gürkaynak, R. S., B. Sack and J. H. Wright (2007), "The U.S. Treasury Yield Curve: 1961 to the Present," Journal of Monetary Economics 54(8), 2291–2304.

Hall, R. E. and D. W. Jorgenson (1967), "Tax Policy and Investment Behavior," American Economic Review 57(3), 391–414.

Hamilton, J. D., E. S. Harris, J. Hatzius and K. D. West (2016), "The Equilibrium Real Funds Rate: Past, Present, and Future," IMF Economic Review 64(4), 660–707.

Hanson, S. G. and J. C. Stein (2015), "Monetary Policy and Long-Term Real Rates," Journal of Financial Economics 115(3), 429–448.

Hatzius, J., P. Hooper, F. Mishkin, K. Schoenholtz and M. Watson (2010), "Financial Conditions Indexes: A Fresh Look after the Financial Crisis," NBER Working Paper 16150; also in U.S. Monetary Policy Forum Report 2010, 3–59.

Haubrich, J., G. Pennacchi and P. Ritchken (2012), "Inflation Expectations, Real Rates, and Risk Premia: Evidence from Inflation Swaps," Review of Financial Studies 25(5), 1588–1629.

Holston, K., T. Laubach and J. C. Williams (2017), "Measuring the Natural Rate of Interest: International Trends and Determinants," Journal of International Economics 108, S59–S75.

Holston, K., T. Laubach and J. C. Williams (2023), "Measuring the Natural Rate of Interest after COVID-19," Federal Reserve Bank of New York Staff Report 1063.

Jacobson, M. M., C. Matthes and T. B. Walker (2026), "Temporal Aggregation Bias and Monetary Policy Transmission," Journal of Political Economy Macroeconomics, doi:10.1086/741548.

Jarociński, M. and P. Karadi (2020), "Deconstructing Monetary Policy Surprises—The Role of Information Shocks," American Economic Journal: Macroeconomics 12(2), 1–43.

Jorgenson, D. W. (1963), "Capital Theory and Investment Behavior," American Economic Review 53(2), 247–259.

Kaplan, G., B. Moll and G. L. Violante (2018), "Monetary Policy According to HANK," American Economic Review 108(3), 697–743.

Kiley, M. T. (2020a), "Financial Conditions and Economic Activity: Insights from Machine Learning," Finance and Economics Discussion Series 2020-095, Board of Governors of the Federal Reserve System.

Kiley, M. T. (2020b), "What Can the Data Tell Us about the Equilibrium Real Interest Rate?" International Journal of Central Banking 16(3), 181–209.

Kilian, L. (2024), "How to Construct Monthly VAR Proxies Based on Daily Surprises in Futures Markets," Journal of Economic Dynamics and Control 168, 104966.

Kuttner, K. N. (2001), "Monetary Policy Surprises and Interest Rates: Evidence from the Fed Funds Futures Market," Journal of Monetary Economics 47(3), 523–544.

Laubach, T. and J. C. Williams (2003), "Measuring the Natural Rate of Interest," Review of Economics and Statistics 85(4), 1063–1070.

Mertens, K. and M. O. Ravn (2013), "The Dynamic Effects of Personal and Corporate Income Tax Changes in the United States," American Economic Review 103(4), 1212–1247.

Miranda-Agrippino, S. and G. Ricco (2021), "The Transmission of Monetary Policy Shocks," American Economic Journal: Macroeconomics 13(3), 74–107.

Mundell, R. A. (1963), "Capital Mobility and Stabilization Policy under Fixed and Flexible Exchange Rates," Canadian Journal of Economics and Political Science 29(4), 475–485.

Nakamura, E. and J. Steinsson (2018), "High-Frequency Identification of Monetary Non-Neutrality: The Information Effect," Quarterly Journal of Economics 133(3), 1283–1330.

Plagborg-Møller, M. and C. K. Wolf (2021), "Local Projections and VARs Estimate the Same Impulse Responses," Econometrica 89(2), 955–980.

Ramey, V. A. (2016), "Macroeconomic Shocks and Their Propagation," in J. B. Taylor and H. Uhlig (eds.), Handbook of Macroeconomics, Vol. 2A, 71–162.

Romer, C. D. and D. H. Romer (2000), "Federal Reserve Information and the Behavior of Interest Rates," American Economic Review 90(3), 429–457.

Sims, C. A. (1992), "Interpreting the Macroeconomic Time Series Facts: The Effects of Monetary Policy," European Economic Review 36(5), 975–1000.

Smets, F. and R. Wouters (2007), "Shocks and Frictions in US Business Cycles: A Bayesian DSGE Approach," American Economic Review 97(3), 586–606.

Stock, J. H. and M. W. Watson (2003), "Forecasting Output and Inflation: The Role of Asset Prices," Journal of Economic Literature 41(3), 788–829.

Stock, J. H. and M. W. Watson (2018), "Identification and Estimation of Dynamic Causal Effects in Macroeconomics Using External Instruments," Economic Journal 128(610), 917–948.

Swanson, E. T. (2021), "Measuring the Effects of Federal Reserve Forward Guidance and Asset Purchases on Financial Markets," Journal of Monetary Economics 118, 32–53.

Tobin, J. (1969), "A General Equilibrium Approach to Monetary Theory," Journal of Money, Credit and Banking 1(1), 15–29.

A Additional tables

Table A1: Responses with industrial production added to the monthly model

		12 months			24 months			36 months		
Step	FCI	3	6	12	3	6	12	3	6	12
A. Unemployment gap (percentage points)										
Nominal	none	0.07	−0.04	−0.19†	0.26†	0.16	−0.04	0.26	0.30†	0.19
	included	0.41**	0.36**	0.22**	0.60**	0.59**	0.47**	0.51**	0.56**	0.49**
Real	none	0.12	0.03	−0.14†	0.24†	0.15	−0.02	0.24†	0.23	0.15
	included	0.41**	0.37**	0.24**	0.54**	0.53**	0.46**	0.43**	0.48**	0.45**
r^*-gap	none	0.16†	0.16†	0.03	0.30**	0.27†	0.14	0.33**	0.30†	0.25†
	included	0.44**	0.42**	0.34**	0.52**	0.50**	0.50**	0.42**	0.42**	0.42**
B. Industrial production (percent)										
Nominal	none	1.06**	1.26**	1.65**	1.03†	1.26†	1.64**	1.16†	1.00†	1.10†
	included	−0.16	−0.29	0.11	−0.09	−0.04	0.15	0.45	0.39	0.43
Real	none	0.64†	0.77**	1.13**	0.72	0.97†	1.28**	0.87†	0.85†	0.91†
	included	−0.33	−0.52	−0.23	−0.11	−0.11	−0.01	0.42	0.31	0.34
r^*-gap	none	0.13	−0.04	0.27	0.13	0.23	0.32	0.31	0.44	0.21
	included	−0.84**	−1.05**	−0.95**	−0.46†	−0.48	−0.80**	0.18	0.08	−0.21

See Table 4. Price-level responses are similar to Table 4. Bauer and Swanson (2023a), industrial production: about −1.5, −1.0 and −0.85 at 12, 24 and 36 months, read from their Figure 8.

Table A2: Other surprise measures, with the FCI

		12 months			24 months			36 months		
Shock	Step	3	6	12	3	6	12	3	6	12
A. Unemployment gap (percentage points)										
Nakamura–Steinsson	Nominal	0.44**	0.36**	0.28**	0.76**	0.75**	0.63**	0.76**	0.83**	0.77**
	r^*-gap	0.40**	0.37**	0.36**	0.47**	0.48**	0.50**	0.35**	0.40**	0.43**
Nakamura–Steinsson, Greenbook part removed	Nominal	0.43**	0.34**	0.27**	0.75**	0.73**	0.62**	0.75**	0.81**	0.76**
	r^*-gap	0.39**	0.36**	0.34**	0.46**	0.47**	0.48**	0.34**	0.39**	0.42**
Bauer–Swanson, predictable part removed	Nominal	0.47**	0.39**	0.32**	0.81**	0.79**	0.67**	0.81**	0.87**	0.80**
	r^*-gap	0.44**	0.41**	0.40**	0.52**	0.52**	0.54**	0.40**	0.44**	0.46**
Jarociński–Karadi pure policy	Nominal	0.48**	0.40**	0.33**	0.82**	0.81**	0.69**	0.83**	0.89**	0.81**
	r^*-gap	0.45**	0.42**	0.41**	0.53**	0.54**	0.55**	0.41**	0.45**	0.47**
B. Core PCE price level (percent)										
Nakamura–Steinsson	Nominal	−0.09	−0.11	−0.11	−0.24	−0.36†	−0.31†	−0.33	−0.54†	−0.49†
	r^*-gap	−0.34**	−0.33**	−0.33**	−0.53**	−0.68**	−0.72**	−0.57**	−0.85**	−0.97**
Nakamura–Steinsson, Greenbook part removed	Nominal	−0.08	−0.11	−0.11	−0.24	−0.35†	−0.30	−0.33	−0.54†	−0.48
	r^*-gap	−0.34**	−0.33**	−0.33**	−0.52**	−0.67**	−0.71**	−0.56**	−0.83**	−0.95**
Bauer–Swanson, predictable part removed	Nominal	−0.09	−0.12	−0.12	−0.25	−0.37†	−0.32	−0.34	−0.55†	−0.51
	r^*-gap	−0.36**	−0.35**	−0.35**	−0.55**	−0.71**	−0.76**	−0.59**	−0.88**	−1.01**
Jarociński–Karadi pure policy	Nominal	−0.09	−0.12	−0.12	−0.25	−0.37†	−0.33	−0.34	−0.56†	−0.51
	r^*-gap	−0.36**	−0.36**	−0.35**	−0.55**	−0.72**	−0.77**	−0.60**	−0.89**	−1.03**

See Table 4. Each surprise is scaled to raise the two-year yield 1 percentage point in the window. Step: the nominal, or the r^*-gap, two-year yield and FCI.

Table A3: Main steps without the time trend

		12 months			24 months			36 months		
Step	FCI	3	6	12	3	6	12	3	6	12
A. Unemployment gap (percentage points)										
Nominal	none	−0.13†	−0.27**	−0.34**	−0.01	−0.19†	−0.36**	0.08	−0.08	−0.25†
	included	0.18**	0.09	0.07	0.38**	0.29**	0.19†	0.42**	0.37**	0.24**
Real	none	−0.02	−0.14†	−0.25**	0.07	−0.10	−0.28**	0.12	−0.03	−0.21†
	included	0.25**	0.18†	0.12	0.40**	0.33**	0.24**	0.38**	0.36**	0.26**
r^*-gap	none	0.08	0.03	−0.08	0.16†	0.04	−0.15†	0.20†	0.05	−0.16
	included	0.30**	0.32**	0.34**	0.38**	0.39**	0.41**	0.34**	0.38**	0.37**
B. Core PCE price level (percent)										
Nominal	none	0.01	0.06	0.01	−0.08	0.03	0.05	−0.17	−0.07	0.03
	included	−0.09	−0.07	−0.10	−0.28†	−0.24†	−0.23	−0.42†	−0.41†	−0.36†
Real	none	−0.10†	−0.02	−0.07	−0.19†	−0.08	−0.06	−0.26†	−0.16	−0.08
	included	−0.17†	−0.14†	−0.18**	−0.33†	−0.32†	−0.31†	−0.44†	−0.47**	−0.41†
r^*-gap	none	−0.15†	−0.13†	−0.19**	−0.22†	−0.26†	−0.31†	−0.28†	−0.39	−0.44†
	included	−0.18†	−0.25**	−0.33**	−0.28†	−0.48**	−0.61**	−0.37†	−0.68**	−0.84**

As Table 4, but without the quadratic time trend.

Table A4: Main steps with a fuller measure of uncertainty

		12 months			24 months			36 months		
Step	FCI	3	6	12	3	6	12	3	6	12
A. Unemployment gap (percentage points)										
Nominal	none	0.07	$-0.13^{\dagger}$	-0.27^{**}	$0.36^{\dagger}$	0.10	$-0.19^{\dagger}$	0.53^{**}	0.29	-0.00
	included	0.44^{**}	0.36^{**}	0.28^{**}	0.76^{**}	0.75^{**}	0.63^{**}	0.76^{**}	0.83^{**}	0.77^{**}
Real	none	$0.14^{\dagger}$	-0.04	-0.20^{**}	0.33^{**}	0.12	$-0.13^{\dagger}$	0.42^{**}	0.23	-0.01
	included	0.44^{**}	0.38^{**}	0.29^{**}	0.65^{**}	0.65^{**}	0.59^{**}	0.57^{**}	0.64^{**}	0.64^{**}
r^*-gap	none	0.19^{**}	0.10	-0.03	0.34^{**}	$0.21^{\dagger}$	0.04	0.38^{**}	$0.25^{\dagger}$	0.07
	included	0.40^{**}	0.37^{**}	0.36^{**}	0.47^{**}	0.48^{**}	0.50^{**}	0.35^{**}	0.40^{**}	0.43^{**}
B. Core PCE price level (percent)										
Nominal	none	-0.03	-0.01	-0.02	-0.16	-0.16	-0.07	-0.26	-0.34	-0.18
	included	-0.09	-0.11	-0.11	-0.24	$-0.36^{\dagger}$	$-0.31^{\dagger}$	-0.33	$-0.54^{\dagger}$	$-0.49^{\dagger}$
Real	none	$-0.14^{\dagger}$	-0.08	-0.10	$-0.25^{\dagger}$	$-0.23^{\dagger}$	-0.16	$-0.33^{\dagger}$	$-0.37^{\dagger}$	-0.26
	included	$-0.17^{\dagger}$	$-0.17^{\dagger}$	$-0.18^{\dagger}$	$-0.30^{\dagger}$	$-0.40^{\dagger}$	$-0.36^{\dagger}$	$-0.35^{\dagger}$	$-0.54^{\dagger}$	$-0.49^{\dagger}$
r^*-gap	none	-0.27^{**}	-0.22^{**}	-0.23^{**}	-0.45^{**}	-0.51^{**}	-0.52^{**}	-0.55^{**}	-0.70^{**}	-0.80^{**}
	included	-0.34^{**}	-0.33^{**}	-0.33^{**}	-0.53^{**}	-0.68^{**}	-0.72^{**}	-0.57^{**}	-0.85^{**}	-0.97^{**}

As Table 4, with 1,000 draws. Each draw resamples the announcements, re-estimates the daily model on resampled 66-day blocks, and rebuilds and re-estimates the monthly data with randomly sign-flipped errors. r^* and the FCI weights are held fixed.

Table A5: Unemployment response with the monthly equations estimated on longer samples

		12 months			24 months			36 months		
Step	FCI	3	6	12	3	6	12	3	6	12
Monthly equations estimated from 1973: Unemployment gap (percentage points)										
Nominal	none	0.02	0.03	0.09	$0.11^{\dagger}$	$0.10^{\dagger}$	0.16	$0.16^{\dagger}$	0.10	0.11
	included	0.09^{**}	0.20^{**}	0.29^{**}	0.13^{**}	0.16^{**}	0.24^{**}	0.14^{**}	$0.09^{\dagger}$	$0.11^{\dagger}$
Real	none	0.02	0.03	0.06	0.05	0.03	0.04	0.07	0.01	−0.01
	included	0.07^{**}	0.18^{**}	0.24^{**}	$0.08^{\dagger}$	$0.10^{\dagger}$	0.15^{**}	$0.08^{\dagger}$	0.02	0.05
r^*-gap	none	$0.06^{\dagger}$	$0.10^{\dagger}$	0.16^{**}	$0.10^{\dagger}$	0.08	$0.13^{\dagger}$	$0.11^{\dagger}$	0.02	0.04
	included	0.11^{**}	0.23^{**}	0.27^{**}	$0.10^{\dagger}$	$0.12^{\dagger}$	0.18^{**}	0.06	0.02	0.08
Monthly equations estimated from 1977: Unemployment gap (percentage points)										
Nominal	none	0.13^{**}	0.14^{**}	0.08	0.31^{**}	0.30^{**}	$0.17^{\dagger}$	0.41^{**}	0.35^{**}	0.14
	included	0.20^{**}	0.27^{**}	0.26^{**}	0.31^{**}	0.33^{**}	0.29^{**}	0.33^{**}	0.28^{**}	0.19^{**}
Real	none	$0.07^{\dagger}$	0.12^{**}	$0.09^{\dagger}$	0.15^{**}	$0.15^{\dagger}$	0.07	0.18^{**}	0.12	−0.03
	included	0.14^{**}	0.24^{**}	0.24^{**}	0.18^{**}	0.19^{**}	0.18^{**}	0.17^{**}	$0.11^{\dagger}$	0.04
r^*-gap	none	0.12^{**}	0.19^{**}	0.18^{**}	0.20^{**}	0.20^{**}	$0.15^{\dagger}$	0.23^{**}	$0.14^{\dagger}$	0.01
	included	0.19^{**}	0.28^{**}	0.28^{**}	0.20^{**}	0.20^{**}	0.20^{**}	0.15^{**}	$0.09^{\dagger}$	0.07
Monthly equations estimated from 1982: Unemployment gap (percentage points)										
Nominal	none	$0.12^{\dagger}$	−0.01	−0.06	0.40^{**}	$0.22^{\dagger}$	0.09	0.58^{**}	0.38^{**}	0.19
	included	0.27^{**}	0.18^{**}	$0.15^{\dagger}$	0.51^{**}	0.39^{**}	0.28^{**}	0.56^{**}	0.43^{**}	0.28^{**}
Real	none	0.15^{**}	0.03	−0.03	0.35^{**}	$0.19^{\dagger}$	0.05	0.46^{**}	$0.28^{\dagger}$	0.07
	included	0.29^{**}	0.21^{**}	0.17^{**}	0.46^{**}	0.35^{**}	0.25^{**}	0.44^{**}	0.35^{**}	$0.21^{\dagger}$
r^*-gap	none	0.21^{**}	$0.13^{\dagger}$	0.09	0.39^{**}	$0.26^{\dagger}$	0.15	0.46^{**}	$0.28^{\dagger}$	0.09
	included	0.29^{**}	0.25^{**}	0.21^{**}	0.38^{**}	0.29^{**}	0.25^{**}	0.31^{**}	0.23^{**}	0.18^{**}
Monthly equations estimated from 1973, no trend: Unemployment gap (percentage points)										
Nominal	none	−0.00	0.03	$0.12^{\dagger}$	$0.08^{\dagger}$	0.07	$0.18^{\dagger}$	0.14^{**}	0.09	$0.13^{\dagger}$
	included	$0.05^{\dagger}$	0.19^{**}	0.32^{**}	$0.07^{\dagger}$	0.14^{**}	0.27^{**}	$0.08^{\dagger}$	0.08	$0.15^{\dagger}$
Real	none	0.02	0.05	$0.10^{\dagger}$	$0.07^{\dagger}$	0.05	0.08	$0.10^{\dagger}$	0.03	0.02
	included	0.07^{**}	0.20^{**}	0.28^{**}	$0.07^{\dagger}$	0.12^{**}	0.20^{**}	$0.07^{\dagger}$	0.05	$0.11^{\dagger}$
r^*-gap	none	$0.06^{\dagger}$	0.11^{**}	0.18^{**}	$0.11^{\dagger}$	$0.10^{\dagger}$	$0.14^{\dagger}$	$0.13^{\dagger}$	0.04	0.04
	included	0.13^{**}	0.26^{**}	0.31^{**}	0.13^{**}	0.18^{**}	0.26^{**}	$0.11^{\dagger}$	$0.10^{\dagger}$	0.20^{**}

As Table 4, but the monthly equations are fitted from the year shown; surprises (1996 on), announcement windows and the daily model (1990 on) are unchanged. Before the daily data begin, monthly averages are used. FCI: the companion note's monthly index for the step. Two-year yield: FRED before 1990, and before June 1976 the Gürkaynak–Sack–Wright fitted two-year zero-coupon yield, shifted by its average gap to the FRED yield over 1976–1989 (about 0.25 percentage point). Expected inflation: Cleveland Fed from 1982; before 1982, the companion note's one-year expectations (Livingston survey based), which is a splice. The 1977 start avoids the yield splice, and the 1982 start avoids both. The last panel drops the time trend.

Table A6: Price-level response with the monthly equations estimated on longer samples

		12 months			24 months			36 months		
Step	FCI	3	6	12	3	6	12	3	6	12
Monthly equations estimated from 1973: Core PCE price level (percent)										
Nominal	none	$0.09^{\dagger}$	$0.09^{\dagger}$	0.08	$0.15^{\dagger}$	0.06	0.04	$0.19^{\dagger}$	0.00	−0.06
	included	$0.08^{\dagger}$	0.07	0.07	$0.13^{\dagger}$	0.03	−0.01	$0.15^{\dagger}$	−0.03	−0.09
Real	none	$-0.05^{\dagger}$	$-0.06^{\dagger}$	-0.13^{**}	$-0.10^{\dagger}$	-0.17^{**}	-0.27^{**}	$-0.14^{\dagger}$	-0.26^{**}	-0.35^{**}
	included	−0.04	$-0.09^{\dagger}$	-0.15^{**}	$-0.09^{\dagger}$	-0.21^{**}	-0.32^{**}	$-0.13^{\dagger}$	-0.30^{**}	-0.40^{**}
r^*-gap	none	$-0.08^{\dagger}$	-0.10^{**}	-0.15^{**}	$-0.16^{\dagger}$	-0.25^{**}	-0.35^{**}	$-0.21^{\dagger}$	-0.36^{**}	-0.48^{**}
	included	$-0.09^{\dagger}$	-0.13^{**}	-0.18^{**}	-0.18^{**}	-0.31^{**}	-0.40^{**}	-0.26^{**}	-0.44^{**}	-0.53^{**}
Monthly equations estimated from 1977: Core PCE price level (percent)										
Nominal	none	0.02	0.02	−0.01	0.02	−0.07	−0.16	0.01	−0.20	$-0.37^{\dagger}$
	included	0.01	−0.00	−0.06	−0.00	−0.10	$-0.23^{\dagger}$	−0.00	−0.20	-0.42^{**}
Real	none	$-0.09^{\dagger}$	-0.09^{**}	-0.15^{**}	-0.18^{**}	-0.24^{**}	-0.37^{**}	-0.24^{**}	-0.38^{**}	-0.56^{**}
	included	-0.09^{**}	-0.11^{**}	-0.19^{**}	-0.18^{**}	-0.26^{**}	-0.43^{**}	-0.24^{**}	-0.38^{**}	-0.62^{**}
r^*-gap	none	-0.13^{**}	-0.13^{**}	-0.19^{**}	-0.24^{**}	-0.32^{**}	-0.46^{**}	-0.31^{**}	-0.47^{**}	-0.70^{**}
	included	-0.14^{**}	-0.15^{**}	-0.24^{**}	-0.26^{**}	-0.35^{**}	-0.54^{**}	-0.34^{**}	-0.52^{**}	-0.81^{**}
Monthly equations estimated from 1982: Core PCE price level (percent)										
Nominal	none	−0.00	0.02	−0.01	−0.02	−0.06	−0.08	−0.04	−0.14	−0.18
	included	−0.02	−0.02	−0.05	−0.05	−0.11	$-0.20^{\dagger}$	−0.07	−0.19	$-0.33^{\dagger}$
Real	none	$-0.07^{\dagger}$	−0.05	$-0.08^{\dagger}$	−0.12	$-0.15^{\dagger}$	−0.16	−0.16	$-0.23^{\dagger}$	−0.24
	included	$-0.08^{\dagger}$	$-0.08^{\dagger}$	$-0.12^{\dagger}$	$-0.14^{\dagger}$	$-0.20^{\dagger}$	-0.28^{**}	$-0.17^{\dagger}$	$-0.28^{\dagger}$	$-0.40^{\dagger}$
r^*-gap	none	-0.15^{**}	-0.14^{**}	-0.16^{**}	-0.25^{**}	-0.32^{**}	-0.39^{**}	-0.32^{**}	-0.45^{**}	-0.59^{**}
	included	-0.17^{**}	-0.17^{**}	-0.22^{**}	-0.28^{**}	-0.39^{**}	-0.51^{**}	-0.32^{**}	-0.52^{**}	-0.74^{**}
Monthly equations estimated from 1973, no trend: Core PCE price level (percent)										
Nominal	none	0.16^{**}	$0.10^{\dagger}$	−0.01	0.44^{**}	$0.20^{\dagger}$	−0.09	0.72^{**}	0.29	−0.16
	included	0.13^{**}	0.06	−0.01	0.29^{**}	0.05	−0.18	0.40^{**}	0.00	$-0.36^{\dagger}$
Real	none	−0.03	$-0.10^{\dagger}$	-0.24^{**}	0.01	$-0.21^{\dagger}$	-0.51^{**}	0.05	$-0.32^{\dagger}$	-0.72^{**}
	included	−0.03	$-0.14^{\dagger}$	-0.26^{**}	−0.06	-0.32^{**}	-0.57^{**}	−0.10	-0.47^{**}	-0.81^{**}
r^*-gap	none	$-0.11^{\dagger}$	-0.17^{**}	-0.29^{**}	−0.20	-0.39^{**}	-0.64^{**}	−0.27	-0.62^{**}	-0.96^{**}
	included	$-0.10^{\dagger}$	-0.20^{**}	-0.32^{**}	$-0.23^{\dagger}$	-0.49^{**}	-0.75^{**}	$-0.39^{\dagger}$	-0.78^{**}	-1.12^{**}

See Table A5.

Table A7: Comparison with purely monthly VARs

		Nominal			Real			r^*-gap		
Model	FCI	3	6	12	3	6	12	3	6	12
A. Unemployment gap after 12 months (percentage points)										
Baseline mixed-frequency model	none	+0.07	−0.13	−0.27**	+0.14	−0.04	−0.20**	+0.19**	+0.10	−0.03
	included	+0.44**	+0.36**	+0.28**	+0.44**	+0.38**	+0.29**	+0.40**	+0.37**	+0.36**
Monthly, window impact, announcement timing	none	+0.06	−0.05	−0.14**	+0.08	−0.01	−0.10	+0.11**	+0.06	+0.00
	included	+0.21**	+0.17**	+0.11	+0.23**	+0.19**	+0.14**	+0.22**	+0.23**	+0.21**
Monthly, window impact, first day of month	none	+0.13	−0.11	−0.30**	+0.17	−0.03	−0.21	+0.25**	+0.14	+0.00
	included	+0.46**	+0.36**	+0.23	+0.49**	+0.42**	+0.30**	+0.48**	+0.51**	+0.47**
Monthly proxy, macro zero on impact	none	+0.13	−0.11	−0.30	+0.17	−0.03	−0.21	+0.25	+0.14	+0.00
	included	+0.06	−0.10°	−0.28°	+0.08	−0.04	−0.12°	+0.15	+0.14	+0.17
Monthly proxy, standard	none	−0.24	−0.44	−0.64	−0.30	−0.51	−0.76**	−0.22	−0.40	−0.68
	included	−0.04	−0.18°	−0.38°	−0.11	−0.19	−0.37°	+0.02	−0.01	−0.11
Monthly proxy, standard, funds rate and MP1	none	−0.37	−0.73	−0.90**	−0.16	−0.42	−0.59**	−0.04	−0.25	−0.43
	included	−0.74**	−0.98**	−1.02**	−0.47	−0.60	−0.62**	−0.39	−0.52	−0.54
B. Core PCE price level after 24 months (percent)										
Baseline mixed-frequency model	none	−0.16	−0.16	−0.07	−0.25	−0.23	−0.16	−0.45**	−0.51**	−0.52**
	included	−0.24	−0.36	−0.31	−0.30	−0.40	−0.36	−0.53**	−0.68**	−0.72**
Monthly, window impact, announcement timing	none	−0.08	−0.08	−0.03	−0.12	−0.12	−0.07	−0.25**	−0.29**	−0.27**
	included	−0.11	−0.15	−0.10	−0.14	−0.19	−0.12	−0.29**	−0.39**	−0.34**
Monthly, window impact, first day of month	none	−0.17	−0.18	−0.07	−0.25	−0.25	−0.15	−0.53**	−0.62**	−0.58**
	included	−0.23	−0.32	−0.22	−0.31	−0.41	−0.27	−0.62**	−0.84**	−0.75**
Monthly proxy, macro zero on impact	none	−0.17	−0.18	−0.07	−0.25	−0.25	−0.15	−0.53	−0.62	−0.58
	included	−0.13	−0.18°	−0.06°	−0.20**	−0.25	−0.17°	−0.41**	−0.57**	−0.54
Monthly proxy, standard	none	−0.27	−0.28	−0.04	−0.31	−0.37	−0.33	−0.61	−0.68	−0.77
	included	−0.35	−0.50°	−0.50°	−0.32	−0.54	−0.83°	−0.57	−0.81**	−1.07**
Monthly proxy, standard, funds rate and MP1	none	−0.15	+0.03	+0.26	−0.18	−0.08	−0.04	−0.50	−0.55	−0.60
	included	+0.06	+0.15	+0.28	−0.03	−0.02	−0.10	−0.13	−0.22	−0.24
C. First-stage F statistic of the monthly proxy										
Two-year yield, NS surprise	none	5.6	5.4	5.5	6.1	5.8	5.0	5.8	6.3	4.9
	included	3.1	2.5	1.7	4.1	3.4	1.7	6.4	5.9	3.2
Funds rate, MP1 surprise	none	7.7	6.5	7.3	8.0	7.5	8.3	6.4	5.8	6.5
	included	7.9	6.7	8.1	7.3	6.9	7.9	5.6	5.3	6.3

Columns: step, then the number of monthly lags. All models include the quadratic time trend. Proxy cases: per percentage point on the monthly average of the two-year yield (funds rate in the last proxy row); first-day case: the same; baseline and announcement-timing case: per percentage point on the two-year yield in the window. **: 90 percent interval or Anderson–Rubin set excludes zero. °: the 90 percent Anderson–Rubin set is unbounded. In the funds-rate rows, the funds rate receives the same adjustment as the two-year yield.

B Data sources

Table A8 lists the data. Series from FRED are identified by their mnemonics. Quarterly series are placed in the middle month of the quarter and interpolated linearly.

Table A8: Data sources

Series	Frequency	Source
Unemployment rate	Monthly	FRED UNRATE
Natural rate of unemployment	Quarterly	CBO, FRED NROU
Core PCE price index	Monthly	FRED PCEPILFE
Two-year Treasury yield	Daily	FRED DGS2
10-year Treasury yield	Daily, monthly	FRED DGS10, GS10
Expected inflation, 2 and 10 years	Monthly	Cleveland Fed, FRED EXPINF2YR, EXPINF10YR
r^*, Laubach–Williams	Quarterly	New York Fed, one-sided estimates
CAPE	Monthly	Shiller, spliced over 1990–92 to the monthly average of daily CAPE
Daily CAPE	Daily	S&P 500 (FRED SP500; before October 2016, Dow Jones S&P 500 history, level-matched) over Shiller's 10-year earnings
Real broad dollar	Monthly	FRED RTWEXBGS, spliced to TWEXBPA in 2006
Nominal broad dollar	Daily	FRED DTWEXBGS, earlier DTWEXB, TWEXBMTH
Futures, yields, S&P 500 and DXY in the window	Event	U.S. Monetary Policy Event-Study Database, "monetary events" (Acosta et al., 2025)

C Relaxing the key assumptions

Section 3 describes two restrictions specific to this note. The surprise reaches unemployment and prices only through the two-year yield and the FCI, and the three parts of the FCI act in proportion to their weights. This appendix relaxes each in turn, and then adds a monthly credit measure to the monthly block. Table A9 repeats the main steps for the first two exercises. In each exercise the daily model and the monthly equations are extended with the new variables. Everything else is as in the baseline, including the quadratic trend and 200 draws.

Adding a credit spread. The daily model and the monthly equations now include the Baa-10-year spread: Moody's Baa corporate yield minus the 10-year Treasury yield, available daily from 1986. It is a rough measure of credit conditions. Unlike the other variables in the model, it can respond to the surprise on impact. It has no 30-minute series, so its impact is the change over the announcement day regressed on the surprise, scaled to a 1 percentage point window move in the two-year yield. The impact is -0.25 percentage point. A tightening narrows this spread on the day, because corporate yields rise less than the 10-year Treasury yield. This is not the widening that a credit channel would suggest. After the announcement the spread moves little: its monthly average response stays between -0.16 and $+0.01$ percentage point over two years.

The effects of the surprise barely change (Table A9). At the r^* step, unemployment rises 0.28 to 0.35 percentage point after 12 months, compared with 0.36 to 0.40 in the baseline, and the price level falls 0.48 to 0.65 percent after 24 months, compared with 0.53 to 0.72. All are significant at the 90 percent level for every lag length. The nominal and real steps are also close to the baseline. So leaving out this spread does not drive the results. The test is limited. The spread is crude, because it includes expected default, and the excess bond premium that purges it is available only monthly. Earlier work in this project found that credit variables matter for the monetary-shock VAR, so a better measure may behave differently.

Entering the parts of the FCI separately. The FCI is replaced by its three weighted parts: the 10-year yield, equity valuation (CAPE) and the real dollar, each in the same step (nominal, real or r^*-gap) as the FCI. The parts sum to the FCI. Each enters the daily and monthly equations with its own coefficients, so their effects on unemployment and prices are no longer tied to the weights. Their impacts per 1 percentage point move in the two-year yield are 0.32, 0.39 and 0.27 index units (together 0.99, as for the FCI).

Signs and broad size agree with the baseline (Table A9), but precision falls. At the r^* step, unemployment rises 0.48, 0.33 and 0.18 percentage point after 12 months with 3, 6 and 12 lags. The first two are significant at the 90 percent level; the third is significant only at 68 percent. The price level falls 0.30 percent after 12 months with 12 lags and 0.50 percent after 24 months, significant at the 90 percent level; with 3 and 6 lags the 24-month falls are 0.24 and 0.41, significant only at the 68 percent level at 6 lags. At the nominal and real steps, unemployment is smaller and less precise at long lags, and prices barely move. The loss of precision is expected. The three parts add $2L$ coefficients to each equation, for 382 months of data.

Table A11 tests the restriction directly. The FCI model is nested in the model with the parts, so an F test applies. It does not reject for unemployment at any step (p-values 0.10 to 0.67). For inflation it is rejected at the 5 percent level at the nominal step with every lag length and at the real step with 12 lags. At the r^* step it is not rejected at 5 percent (p-values 0.08 to 0.24). The tests are plain F tests with many restrictions and no correction for serial correlation, so they are indicative. Taken together, the FCI weights, which come from the response to policy news, summarize how unemployment responds well. For inflation they are a worse summary when rates are not measured relative to r^*, which fits the main result that the r^* step matters most for prices.

Adding the monthly excess bond premium. The excess bond premium (Gilchrist and Zakrajšek, 2012) is the part of corporate bond spreads not explained by default risk. It is monthly, so it enters the monthly block as a third variable, treated like unemployment and inflation. It depends on its own lags and on lagged monthly averages of the two-year yield and the FCI. Within a month, it is set before the financial markets move, so the daily model responds to its current and lagged values. Its response to the surprise is not set to anything: as for unemployment and inflation, it is zero in the announcement window and then evolves with the model. Table A10 gives the results.

Without the FCI, the premium removes the wrong-signed unemployment response. At the nominal step, unemployment at 12 months is -0.27 with 12 lags in the baseline and $+0.17$ with the premium; at the real step it goes from -0.20 to $+0.06$. With the FCI, adding the premium changes little. At the r^* step, unemployment at 12 months is 0.39 to 0.41, compared with 0.36 to 0.40, and the price level at 24 months falls 0.52 to 0.71 percent, compared with 0.53 to 0.72. The prices result at the r^* step without the FCI is also unchanged. The premium itself barely responds to the surprise: its response is within 0.13 percentage point at 12 months in every case, and is not significant with the FCI. So the FCI and the premium are substitutes for fixing the unemployment puzzle, but the FCI does not depend on the premium. This matches the earlier finding from the monetary-shock VAR of this project, where credit variables, not the market index, removed the wrong-signed post-1994 response.

Table A9: Relaxing the key assumptions: credit spread added, and the FCI replaced by its three parts

		12 months			24 months			36 months		
Step	Financial block	3	6	12	3	6	12	3	6	12
A. Unemployment gap (percentage points)										
Nominal	FCI (baseline)	0.44**	0.36**	0.28**	0.76**	0.75**	0.63**	0.76**	0.83**	0.77**
	FCI and Baa spread	0.36**	0.32**	0.25**	0.55**	0.59**	0.62**	0.50**	0.58**	0.68**
	Components instead of FCI	0.37**	0.24†	0.08	0.51**	0.43**	0.21†	0.45**	0.42**	0.31**
Real	FCI (baseline)	0.44**	0.38**	0.29**	0.65**	0.65**	0.59**	0.57**	0.64**	0.64**
	FCI and Baa spread	0.35**	0.30**	0.21†	0.46**	0.48**	0.51**	0.35**	0.41**	0.49**
	Components instead of FCI	0.37**	0.25**	0.07	0.44**	0.37**	0.19	0.37**	0.34**	0.27†
r^*-gap	FCI (baseline)	0.40**	0.37**	0.36**	0.47**	0.48**	0.50**	0.35**	0.40**	0.43**
	FCI and Baa spread	0.35**	0.34**	0.28**	0.36**	0.37**	0.41**	0.21**	0.24**	0.28**
	Components instead of FCI	0.48**	0.33**	0.18†	0.49**	0.46**	0.31†	0.28**	0.33**	0.29†
B. Core PCE price level (percent)										
Nominal	FCI (baseline)	−0.09	−0.11	−0.11	−0.24	−0.36†	−0.31†	−0.33	−0.54†	−0.49†
	FCI and Baa spread	−0.10	−0.10	−0.13	−0.24†	−0.38†	−0.39†	−0.31†	−0.59†	−0.66†
	Components instead of FCI	0.01	−0.04	−0.11	0.00	−0.06	−0.10	−0.04	−0.11	−0.17
Real	FCI (baseline)	−0.17†	−0.17†	−0.18†	−0.30†	−0.40†	−0.36†	−0.35†	−0.54†	−0.49†
	FCI and Baa spread	−0.14†	−0.12	−0.13	−0.26†	−0.35†	−0.31†	−0.32†	−0.51†	−0.47†
	Components instead of FCI	−0.07	−0.07	−0.13†	−0.08	−0.09	−0.09	−0.13	−0.14	−0.11
r^*-gap	FCI (baseline)	−0.34**	−0.33**	−0.33**	−0.53**	−0.68**	−0.72**	−0.57**	−0.85**	−0.97**
	FCI and Baa spread	−0.31**	−0.28**	−0.28**	−0.48**	−0.61**	−0.65**	−0.54**	−0.79**	−0.89**
	Components instead of FCI	−0.22†	−0.25†	−0.30**	−0.24	−0.41†	−0.50**	−0.23	−0.44†	−0.59**

See Table 4. "FCI and Baa spread": the daily Baa-10-year spread is added and responds on impact (event-day change). "Components instead of FCI": the 10-year yield, equity valuation and real dollar enter separately, each weighted as in the FCI and in the same step.

Table A10: Excess bond premium added to the monthly block

		12 months			24 months			36 months		
Step	Model	3	6	12	3	6	12	3	6	12
A. Unemployment gap (percentage points)										
Nominal	No FCI	0.07	−0.13	−0.27**	0.36†	0.10	−0.19	0.53**	0.29	−0.00
	No FCI, with EBP	0.30**	0.22†	0.17†	0.62**	0.52**	0.48**	0.73**	0.60**	0.75**
	FCI	0.44**	0.36**	0.28**	0.76**	0.75**	0.63**	0.76**	0.83**	0.77**
	FCI, with EBP	0.45**	0.41**	0.37**	0.75**	0.70**	0.74**	0.81**	0.73**	0.89**
Real	No FCI	0.14†	−0.04	−0.20**	0.33**	0.12	−0.13	0.42**	0.23	−0.01
	No FCI, with EBP	0.20**	0.13†	0.06	0.36**	0.26†	0.22	0.41†	0.29†	0.42†
	FCI	0.44**	0.38**	0.29**	0.65**	0.65**	0.59**	0.57**	0.64**	0.64**
	FCI, with EBP	0.35**	0.29**	0.22†	0.48**	0.41†	0.46**	0.44†	0.38†	0.55**
r^*-gap	No FCI	0.19**	0.10	−0.03	0.34**	0.21	0.04	0.38**	0.25†	0.07
	No FCI, with EBP	0.25**	0.21**	0.17†	0.35**	0.28**	0.31**	0.31†	0.20	0.38**
	FCI	0.40**	0.37**	0.36**	0.47**	0.48**	0.50**	0.35**	0.40**	0.43**
	FCI, with EBP	0.39**	0.41**	0.39**	0.46**	0.46**	0.60**	0.38**	0.35**	0.51**
B. Core PCE price level (percent)										
Nominal	No FCI	−0.03	−0.01	−0.02	−0.16	−0.16	−0.07	−0.26†	−0.34	−0.18
	No FCI, with EBP	−0.08	−0.07	−0.04	−0.22†	−0.23	−0.09	−0.32†	−0.37	−0.19
	FCI	−0.09	−0.11	−0.11	−0.24	−0.36†	−0.31†	−0.33	−0.54†	−0.49†
	FCI, with EBP	−0.08	−0.10	−0.14†	−0.22	−0.33	−0.36†	−0.30	−0.51†	−0.62†
Real	No FCI	−0.14†	−0.08	−0.10	−0.25†	−0.23	−0.16	−0.33†	−0.37†	−0.26
	No FCI, with EBP	−0.16**	−0.08	−0.07	−0.26**	−0.24†	−0.13	−0.32†	−0.38†	−0.22
	FCI	−0.17†	−0.17†	−0.18†	−0.30†	−0.40†	−0.36†	−0.35†	−0.54†	−0.49†
	FCI, with EBP	−0.14†	−0.10	−0.14†	−0.24†	−0.31†	−0.31†	−0.29†	−0.47†	−0.49†
r^*-gap	No FCI	−0.27**	−0.22**	−0.23**	−0.45**	−0.51**	−0.52**	−0.55**	−0.70**	−0.80**
	No FCI, with EBP	−0.28**	−0.21**	−0.24**	−0.45**	−0.49**	−0.53**	−0.52**	−0.67**	−0.80**
	FCI	−0.34**	−0.33**	−0.33**	−0.53**	−0.68**	−0.72**	−0.57**	−0.85**	−0.97**
	FCI, with EBP	−0.34**	−0.29**	−0.32**	−0.52**	−0.61**	−0.71**	−0.58**	−0.77**	−0.97**
C. Excess bond premium (percentage points)										
Nominal	No FCI	0.13†	0.12†	0.07	0.07	0.03	0.19**	−0.02	−0.03	0.10†
	FCI	0.06	0.04	0.05	0.01	−0.04	0.05	−0.01	−0.03	0.00
Real	No FCI	0.03	0.00	−0.08†	−0.00	−0.02	0.10†	−0.04†	−0.05	0.08
	FCI	−0.02	−0.06	−0.07	−0.05	−0.07	0.00	−0.03	−0.02	0.01
r^*-gap	No FCI	0.02	−0.00	−0.10	−0.04	−0.07†	0.06	−0.07†	−0.08**	0.02
	FCI	0.02	−0.00	0.04	−0.04	−0.07	−0.00	−0.02	−0.02	−0.00

See Table 4. "With EBP": the monthly excess bond premium is a third monthly variable. Panel C is its response to the surprise.

Table A11: Tests of the proportional-weights restriction: p-values

Step	Equation	3 lags	6 lags	12 lags
Nominal	Unemployment gap	0.58	0.54	0.32
	Core PCE inflation	0.04	0.02	0.03
Real	Unemployment gap	0.50	0.67	0.33
	Core PCE inflation	0.12	0.06	0.01
r^*-gap	Unemployment gap	0.10	0.17	0.17
	Core PCE inflation	0.12	0.24	0.08

F tests that the three parts of the FCI have equal coefficients in each monthly equation, with the same sample and trend as the baseline. Number of restrictions: $2L$.